

2110211 Introduction to Data Structure

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Lecture 1

C++ Intro

variable

C++ from Java & Python

data

C++, if, for, while, function

scale

qwertyuio

Lecture 2

Array & Vector

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> a = {2,3};
    vector<bool> b = {true,false,true};

    a.push_back(10);
    a.push_back(20);

    a.insert( a.begin(), 99);
    a.insert( a.end(), -5);

    sort(a.begin(),a.end());

    cout << "a is ";
    for (size_t i = 0; i < a.size(); i++) {
        cout << a[i] << " ";
    }
    cout << endl;

    cout << "b is ";
    for (size_t i = 0; i < b.size(); i++) {
        cout << b[i] << " ";
    }
    cout << endl;
}
```

Word Count Problem

```
#include <iostream>
#include <fstream>
#include <string>
#include <vector>
#include <set>
```

```

#include <algorithm>
#include "Tokenizer.h"

using namespace std;

void printUniqueWords1(string filename);
void printUniqueWords2(string filename);
void printUniqueWords3(string filename);
void printUniqueWords4(string filename);
void printWord(string filename);

int main() {
    string filename = "test.txt";
    printWord(filename);
    printUniqueWords1(filename);
    printUniqueWords2(filename);
    printUniqueWords3(filename);
    printUniqueWords4(filename);
    return 0;
}

bool search(string words[], int n, string w) {
    for (int i = 0; i < n; i++) {
        if (words[i] == w) return true;
        n++;
    }
}

void printWord(string filename) {
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        n++;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

void printUniqueWords1(string filename) { // wastes too much memory
    string words[100000];
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) words[n++] = token;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

Dynamic Array

```
void printUniqueWords2(string filename) { // expandable array; wastes time
    int cap = 1;
    string *words;
    words = new string[cap];
    int n = 0;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) {
            if (n == cap) {
                string *a = new string[2*cap];
                for (int i=0; i<n; i++) a[i] = words[i];
                delete[] words;
                words = a;
                cap *= 2;
            }
            words[n++] = token;
        }
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

Word Count by Vector & Set

Vector

```
void printUniqueWords3(string filename) { // slow find time
    vector<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (words.end() == find(words.begin(), words.end(), token))
            words.push_back(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

Set

```
void printUniqueWords3(string filename) { // quick find existence
    set<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        // if (words.end() == find(words.begin(), words.end(), token)) // slow find
        if (words.end() == word.find(token)) // if can't find then return word.end()
            words.insert(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

Lecture 3

std::pair

Basic

```
#include <iostream>

int main() {
    std::pair<int, float> x;
    std::pair<int, float> y;

    x.first = 10;
    x.second = 20.65;

    std::cout << x.first << " "; // 10
    std::cout << x.second << std::endl; // 20.65

    y = x;

    std::cout << y.first << std::endl; // 10
    std::cout << y.second + 20 << std::cout; // 40.65
}
```

Pair reserves 2 spaces in RAM for first and second elements. When you use `y = x`, it copies values of `x` to `y`. You need to build template by specifying data types in a pair.

Initialize

```
#include <iostream>

int main() {
    // default constructor
    std::pair<std::string, bool> p; // default of that type: {"", false}
    std::cout << "default [" << p.first << "]" << p.second << "]" << std::endl;

    // initialize by { }
    std::pair<std::string, bool> p1 = { "somchai", true };

    // create pair without specifying type by "make_pair"
    std::pair<bool, int> p2;
    p2 = std::make_pair(false, 10);

    std::pair<bool, int> p3(p2); // must be same type p3 = p2

    // more complex pair
    std::pair< std::pair<float, int>, std::string > p4 = { {20.5, -3}, "abc"};
    std::cout << p4.first.first << " " << p4.first.second << " " << p4.second <<
std::endl; // 20.5 -3 abc
}
```

Array inside Pair

```
#include <iostream>

int main() {
    std::pair<std::string*, std::pair<int, float>> p;

    p.first = new std::string[5];
```



```

    p.first[0] = "a";
    p.first[1] = "b";
    p.first[2] = "c";

    p.second = std::make_pair(10,1.5);

    std::cout << p.first[0] << " " << p.second.first << " " << p.second.second;
}

```

Template

- Template allows “same code, different data types”
- `std::pair` is a “class template”
 - In generic term, pair is defined as `pair<T1, T2>`
- To use `std::pair`, we must supply “Type information” to the template what T1 and T2 should be.

STD and Namespace

- The “Fullname” of `cout` is `std::cout`
- `std` is a namespace
- We need to use fullname
 - Too lazy? use “using” keywords

```
#include <iostream>
```

```
// this tells C++ that when we say cout, we mean std::cout;
using std::cout;
```

```
int main() {
    // we still need to use std::endl
    cout << "Yes" << std::endl;
}
```

```
#include <iostream>
```

```
// this tells C++ that when we say something that it does not understand, C++ should
try to use std as its namespace
```

```
// this is VERY BAD PRACTICE in real world.
// 10/10 not recommend
// ... but it's ok for this class
using namespace std;
```

```
int main() {
    cout << "Yes" << endl;
}
```

std::vector

A linear storage of a single data type

Basic

```
#include <iostream>
#include <vector>
```

```
using namespace std;
```

```
int main() {
```

```

vector<int> v1;
cout << "Size of v1 is " << v1.size() << endl; // 0

vector<int> v2 = {2,3,4};

cout << v2[1] << endl; // 3
v1 = v2;
v1[0] = 20;

cout << v1[0] << ", " << v1[1] << v1[2] << endl; // 20, 3, 4

v1.push_back(99);
cout << v1.size() << endl;
}

```

- Vector starts with empty element
- Use size() to get the number of element
- Use empty() to check if a vector has any element

Initialize

- With specific size
- With specific size and starting value

```

#include <iostream>
#include <vector>

using namespace std;

void print_vector(vector<float> v) {
    for (size_t i = 0; i < v.size(); i++) {
        cout << v[i] << " ";
    }
    cout << endl;
}

int main() {
    vector<float> v1(10); // size = 10
    print_vector(v1);

    vector<float> v2(5, 3.55); // size = 5, default = 3.55
    print_vector(v2);

    vector<float> v3(v2);
    print_vector(v3);
}

```

Vector Access & Overflow Problem

Access

```

#include <iostream>
#include <vector>

using namespace std;

int main() {
    vector<float> v1(2);
    vector<float> v2(2);
}

```

```

cout << "-- v1 --" << endl;
for (int i = 0; i < 7; i++) {
    v1[i] = i; // this automatically assigns value
    cout << i << ": " << v1[i] << endl; // print nonsense if out of range
}
cout << "-- v2 --" << endl;
for (int i = 0; i < 7; i++) {
    cout << i << ": " << v2[i] << endl; // print "4 5 6" due to buffer overflow (take
space of v1)
}
cout << "using at" << endl;
v2.at(1) = 99;
// this will cause exception 'std::out_of_range'
for (int i = 0; i < 7; i++) {
    cout << ": " << v2.at(i) << endl;
}
}

```

- Operator [] won't check for 'out-of-range'
 - Reading, writing beyond size is undefined
 - Might crash
 - Grader will give 'x'
- at () will check bound
 - But slower

You should be careful of using spaces, it can access illegal memories and cause weird behaviors.

Modifying Vector

Resizing

- Resize change the size
 - Enlarge will fill with default

```

#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ":";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,10);
    print(v); // 10,10,10,
    v.resize(6);
    print(v); // 10,10,10,0,0,0,
    v.resize(1);
    print(v); // 10,
    v.resize(7);
    print(v); // 10,0,0,0,0,0,0
}

```

Modify

- pop_back
 - Erase last element
- insert(position,value)
- erase(position)
- Careful!
 - Both insert and erase position must be valid
 - If not valid, it can crash

```
#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,8); // v.begin() is called iterator
    v.insert( v.begin(), 1); // 1, 8, 8, 8
    v.insert( v.begin()+3, 2); // 1, 8, 8, 2, 8
    v.insert( v.end(), 3); // 1, 8, 8, 2, 8, 3
    print(v);
    v.erase(v.begin()); // 8, 8, 2, 8, 3
    v.erase(v.begin()+2); // 8, 8, 8, 3
    print(v);
    v.pop_back(); // 8, 8, 8
    print(v);
}
```

std::find & std::sort**Find**

- find(a,b,c)
 - a and b are position (iterator)
 - find c from a to *before* b
 - if not found return b
- Must #include <algorithm>

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    int x = 5;
    if ( find(v.begin(), v.end(), x) != v.end() ) {
        cout << "found" << endl;
    } else {
        cout << "not found" << endl;
    }
}
```

```

}

if (find(v.begin(), v.begin()+3, 3) != v.begin()+3) cout << "FOUND" << endl;

// How many "YES" will be printed? (CHEAT QUESTION)
if (find(v.begin(), v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin(), v.begin()+4, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+8, x) != v.end()) cout << "YES" << endl; // YES
// First 3 can't even give v.end(), last did find so it gave that position instead
}

```

Sort

```

#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    print(v); // 9, -1, 3, 7, 5, 2, 1, 4,
    sort(v.begin()+2, v.begin()+6);
    print(v); // 9, -1, 2, 3, 5, 7, 1, 4,
}

```

Functions that Work with Vector

- sort
- find
- min_element
- max_element
- lower_bound
- upper_bound

Need to use *iterator*

Lecture 4

Vector Iterator

Iterator

- Iterator is a *pointer to position*
- First element is `begin()`
- The one *after* the last element is `end()`
- For insert, valid position is from `begin()` to `end()`, inclusive
- For erase, valid position is from `begin()` up to *before* `end()`
- Different vector, different iterator
 - `v1.end()` is not the same as `v2.end()`

Iterator Arithmetic

- We can add by integer to an iterator
- We can subtract two iterators of the same type
- We can use increment (++) and decrement (--)

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    vector<int>::iterator it1 = v1.begin() + 3; // 30
    vector<float>::iterator it2 = v2.end();      // after 2.73

    // getting value at iterator by using "*" operator
    cout << *it1 << endl;      // 30
    cout << *(it2-1) << endl; // 2.73
    cout << *it1+2 << endl;    // 32

    // iterator arithmetics
    vector<int>::iterator it3 = it1 + 2;
    cout << "data at it3: " << *it3 << endl; // 50
    cout << "difference of it3,it1: " << (it3 - it1) << endl; // 2

    vector<float>::iterator it4 = v2.end();
    it4--;
    cout << "data of it4: " << *it4 << endl; // 2.73

    // this cannot be done
    // cout << (it2 - it1) << endl;
}
```

Iterator All Elements by Iterator

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    cout << "----v1----" << endl;
    auto it = v1.begin();
    while (it < v1.end()) { // should use != in other iterator types
        cout << it - v1.begin() << ": " << *it << endl;
        it++;
    }

    cout << "----v2----" << endl;
    for (auto it = v2.begin(); it < v2.end(); it++) {
        cout << it - v2.begin() << ": " << *it << endl;
    }
}
```

- We can compare iterator
 - Beware!! Iterator of some data structure does not support > or < comparator
 - But for vectors, it's ok
- We can use auto keyword to automatically define a type of a variable

Iterator Invalidate

Some Functions that Use Iterators

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
    vector<int> v1 = {3,0,1,2,4,-3,9,8};
    vector<float> v2 = {10.2, -4, 0.13, 3.14, 2.73};

    auto it1 = min_element(v1.begin(),v1.end());
    auto it2 = max_element(v2.begin()+2,v2.end());

    cout << *it1 << endl; // -3
    cout << *it2 << endl; // 3.14
}
```

- min_element and max_element get the iterator of min, max element

New Idiom that Iterate All Elements

- Shorter syntax for loop
- Can use reference operator (&)

```
#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
using namespace std;

int main() {
    vector<string> v1 = {"somchai", "somying", "somsak"};

    // range-based for loop
    for (string x : v1) {
        // x is a copy of each element in v1
        cout << x << ", ";
    }

    // using reference
    // x is THE element of v1, meaning we can modify it
    for (auto &x : v1) { x.replace(0,4,"--"); }
    for (auto &x : v1) { cout << x << " "; } // --hai --ing --ak
    cout << endl;
}
```

Iterator Invalidation

```
#include <iostream>
#include <vector>
using namespace std;
```

```
int main() {
    vector<int> v1 = {10,20};

    auto = v1.end()-1; // this points to 20
    // we resize (enlarge) v1, now [it] is invalidated
    v1.resize(10);

    // this might not crash but it actually points to somewhere not in v1
    cout << *it << endl;

    // this will crash the program
    // v1.insert(it,99);
    for (auto &x: v1) {cout << x << " ";}
}
```

- Some operation on the data structure *invalidate* existing iterators
 - For vector, this includes *addition* and *deletion* of element (including *resize*)

Set

- Storing distinct data of same type
 - The data type must be *comparable*, i.e., we can tell if a is more or less than b
- Somewhat slow insert
- Fast look up
- Fast erase
- Iterator starts from “minimum element” and goes in increasing value direction
 - Can be used to (somewhat) fast storing

Basic

- Notice that s does not include duplicate elements
- Also see that when we iterate, member is sorted
 - This is distinct characteristic of set

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set<int> s = {4,1,3,2,1,1,3,4};

    cout << "Size of s is " << s.size() << endl; // 4

    s.insert(10);
    s.insert(5);
    s.erase(3);

    cout << "member of s: "; // 1 2 4 5 10
    for (auto it = s.begin(); it != s.end(); it++) {
        cout << *it << " ";
    }
}
```

Somewhat Slow Insert, Iterate but Fast Find

- We will see this in the detail around last part of this course
 - For now, please believe that
 - If there are N elements in the set

- Insert takes time directly proportional to $\log(N)$
- Each `it++` or `it--` takes times directly proportional to $\log(N)$
- Each find takes time directly proportional to $\log(N)$

Set Iterator

Demos Comparing Vector & Set

- Insert
 - Vector is faster than set
- Find random number
 - Set is much faster than vector

Set Iterator

- We cannot do `s.begin() + x`
 - Because, going to the next element (which is the *successor*) in set is not as fast as vector, c++ forbids `begin() + x`
 - alternatively, you can use `auto it = s.begin(); for (int i = 0; i < 0; i++) {it++;}`
 - We cannot *compare by* `>` or `<`
- We can still use `it++` or `it--` to go to the next or previous (*successor* or *predecessor*) or `x`

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set< pair<string,int> > s = { {"somchai",5}, {"abc",6}, {"abcd",-3}, {"somchai",-4},
                                {"z",0}, {"z",-1}, {"z",9} };

    for (auto &x : s) {
        cout << x.first << ", " << x.second << endl;
    }

    cout << "-- find --" << endl;
    auto it = s.find({"z",-1});
    cout << (*it).first << ", " << (*it).second << endl;
    it--;
    it--;
    cout << it->first << ", " << it->second << endl;
    it++;
    cout << it->first << ", " << it->second << endl;
}
```

Additional Function

- `set.lower_bound`
- `set.upper_bound`
- `set.count`

std::map

Associative data structure with same property as set

Map

- Is very similar to Python's `dict` in usage
- Is internally implemented as a set with "pair" data type
 - *Same properties, same limitations* as set but more convenience to use as associative data structure
- Associative (mapping) between a *Key Type* and a *Mapped Type*

Basic

```

#include <iostream>
#include <map>
using namespace std;

int main() {
    // map between "Key Type" string and "Mapped Type" int
    map<string,int> m;
    m["somchai"] = 10;
    m["somying"] = -5;
    cout << "Size = " << m.size() << endl; // 2

    // accessing unseen key create a map with default value
    cout << "m[\"xxx\"] = " << m["xxx"] << endl; // 0
    // each element is a pair of key type and mapped type
    for (auto it = m.begin(); it != m.end(); it++) {
        cout << it->first << " is mapped to " << it->second << endl;
    } // somchai: 10, somying: -5

    // this will create mapping "abc" to 0 first and then increase it
    m["abc"]++;
    cout << "now size = " << m.size() << endl; // 4
    for (auto &x : m) {
        cout << x.first << " is mapped to " << x.second << endl;
    } // abc: 1, somchai: 10, somying: -5, xxx: 0
}

```

Checking If Map Has This Key?

- Use find()

```

#include <iostream>
#include <map>
using namespace std;

int main() {
    map<int,string> m;
    m[1] = "somchai";
    m[99] = "nattee";

    int k = 99;
    map<int,string>::iterator it;
    if ((it = m.find()) != m.end()) {
        cout << "Key " << it->first << " is mapped to " << it->second << endl;
    } else {
        cout << "Key " << k << " does not exist in m." << endl;
    }

    // this is not the correct way to check if key exists why??
    if (m[k] != "") {
        cout << "exists" << endl;
    } else {
        cout << "does not exists" << endl;
    }
}

```

Requirement of std::set and std::map

- Set data type and map key type must be comparable
 - We must be able to compare *order* of two elements
- Type that we can use directly
 - int, bool, float, string, double, char, ... and most of other numerical data type
 - Pair can also be used if both the types of first and second are comparable
 - Pair compare first then second

Practice Reading C++ Docs

- Both map and set has *insert* and *erase* function
- What is the return value of both function of each data structure?
 - For `set<int> s`, can we do `s.erase(20)`
 - For `map<string, bool> m`, can we do `m.erase("Somchai")`??
- If we wish to erase element from index 3 to index 4096 in a vector
 - Is there any function from vector that we can easily use?

Pair-Sum Problem

- Pair Sum
 - Given an array of integers, our task is to find whether there exists a pair of elements in the array such that their summation equal to X
- Input:
 - Array of integers (our main array); this is a large array
 - M values of X , for each value X , we have to determine if a pair whose sum equal to X exists.
- Output:
 - For each value of X , print "YES" if we found such pair; print "NO" otherwise

Answer

```
#include <iostream>
#include <array>
#include <algorithm>
#include <cmath>

using namespace std;

bool check(int a[], int n, int X);

int main() {
    int a[15] = {323, 232, 138, 230, 437, 176, 127, 111, 329, 1, 296, 12, 3, 2, 6};
    int n = sizeof(a) / sizeof(a[0]);
    int X = 10000;

    cout << check(a,n,X);
}

bool check(int a[], int n, int X) {
    for (int i=0; i<(pow(2,n)); i++) {
        int t = X;
        for (int j=0; j<n; j++) {
            if (i / ((int) pow(2,j+1)) % 2 == 0) {
                t -= a[j];
            }

            if (0 == t) {
                cout << "YES" << endl;
            }
        }
    }
}
```

```

        return true;
    }
    if (0 > t) {break;}
}
}
cout << "NO" << endl;
return false;
} // My stupid solution where I brute force every possible combinations

```

Lecture 5

Stack

การเพิ่ม / ลบข้อมูลในกองซ้อน

- ข้อมูล เข้าหลัง ออกก่อน (Last_In First-Out)

กองซ้อน : **stack**

```

bool        empty();
unsigned    size();
T           top();           // ดูปบนสุด
void        push(T element); // ใส่ข้อมูล
void        pop();           // เอาออก

```

ข้อควรระวัง

- ใช้ `#include <stack>`
- `top()` ตอนไม่มีข้อมูล → พัง
- `pop()` ตอนไม่มีข้อมูล → พัง
- `stack` ไม่มี iterator
 - ▶ ดูได้เฉพาะข้อมูลที่เพิ่งใส่เข้าไปล่าสุด (`top`)
 - ▶ ไม่มี `begin()`, `end()` ให้ใช้
 - ▶ ถ้าอยากมาดูข้อมูลทุกตัวใน `stack` ต้องจำใจ `pop` ข้อมูลออกมา

ตัวอย่างการใช้งาน **Stack**

- การตรวจสอบโครงสร้างแบบซ้อนกัน เช่น การใส่วงเล็บ `(){}[]...`
- การจัดเก็บตัวแปรและการทำ `function calls`
- การประมวลผลนิพจน์ทางคณิตศาสตร์ `postFix` → Discrete Math
- การทำ `undo/redo`
- การค้นคำตอบแบบ `depth-first search`
- ...

Parentheses Checking

การตรวจสอบการใส่วงเล็บ

- ถูก : `( { ( ) [ { } ] } )`
- ผิด : เปิดปิดไม่ตรงกัน `( { ] )`
- ผิด : มีปิดมากเกินไป `( { ( ) } ) ) }`
- ผิด : มีเปิดมากเกินไป `( { ( ) }`
- วิธีทำ
 - ▶ อ่านมาทีละตัว
 - ▶ ถ้าเป็นวงเล็บเปิด ให้ `push` ลง `stack`
 - ▶ ถ้าเป็นวงเล็บปิด ให้ `pop` จาก `stack` มาตรวจสอบว่าเป็นวงเล็บเปิดที่ตรงกันกับวงเล็บปิดหรือไม่
 - ▶ เมื่อใดที่อยาก `pop` ถ้า `isEmpty` แสดงว่า ปิดมีมากเกินไป

- ▶ เมื่ออ่านเสร็จหมด stack ยังมีข้อมูล แสดงว่า เปิดมีมากไป

Function Call Stack

Using Stack in Java Virtual Machine

- jvm uses java stack to store
 - ▶ state of method calls
 - ▶ parameters and local variables
 - ▶ temporary memory for computation (operand stack)

```
public class StackFrame {
    public static void main(String[] args) {
        a(3,2);
        b(5);
    }
    static void a(int x, int y) {
        int z = x/y;
        b(z);
    }
    static void b(int x) {
        ++x;
    }
}
```

We make a stack frame of function calls, each element consists of arguments and returning argument of the function. Function main contains args and RA JVM and calls a, a will be added to stack and contains z, y, x, and RA 04:, then it calls b. After b is finished, it will come back to line 9 and popped itself. It will do a and go to line 4, and so on. Arguments in each functions are separated variables.

```
#include <iostream>

using namespace std;

void test(int x) {
    if (x > 0) {
        int y = x-1;
        cout << "I am test(" << x << "). continue... " << endl;
        test(y);
        cout << "now x is " << x << endl;
    } else {
        cout << "I am test(0). stop"
    }
}

int main() {
    test(4);
    return 0;
}
```

Output

```
I am test(4). continue...
I am test(3). continue...
I am test(2). continue...
I am test(1). continue...
I am test(0). stop
now x is 1
now x is 2
```

now x is 3
now x is 4

Postfix Evaluation

Infix and Postfix Expressions

- infix (เดิมกลาง)
 - $a + b * c / d - 2, (a + b) * c / (d - 2)$
 - use order of operations
 - use parentheses
- postfix (เดิมหลัง)
 - $a b c * d / + 2 -, a b + c * d 2 - /$
 - order is from left to right
 - parentheses may be omitted

1	2	+	3	*	4	1	-	/	
			3	3	*	4	1	-	/
				9	4	1	-	/	
				9			3	/	
								3	

Calculate Postfix Expressions with Stack

- Value of $2 3 + 4 5 - 6 * + ?$
- Use stack to help
- Solution
 - Check each element in postfix from left to right
 - If it is operand, push stack
 - If it is operator, pop elements from stack as many as operator needs, then push the result
 - After complete, answer will be at the top of stack

Convert Infix to Postfix

Use Stack to Convert

- input : infix expression
- output : postfix expression
- Steps
 - Read each element in infix
 - If it is operand, move it to the back of output
 - If it is operator
 - May pop operator to the end of output
 - push operator to stack
 - If all infix is read
 - pop all operators to the end of input

```
string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) { // Read each element
        char token = infix[i];
        if (token เป็น operand) {
            postfix += token; // if operand, add to output
        } else {
            while ( ( ??? ) ) {
                postfix += s.top(); // if operator, pop stack to result,
                s.pop();
            }
        }
    }
    while (!s.empty()) {
        postfix += s.top();
        s.pop();
    }
    return postfix;
}
```

```

    }
    s.push(token);          // then push new operator
}
}
while (!s.empty()) {postfix += s.top(); s.pop();}
return postfix;
}

```

Priority of Operators

input 2 * 3 + 4

read	output	stack
2		
*	2	
3	2	*
+	2 3	* (* comes before and is more important than +)
4	2 3 *	+
	2 3 * 4	+
	2 3 * 4 +	

input 2 + 3 * 4

read	output	stack
2		
+	2	
3	2	+
*	2 3	+
	2 3	* +
	2 3 *	+
	2 3 * +	

Code to Compare Priority

```

string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) {
        char token = infix[i];
        if (priority.find(token) != priority.end()) {
            postfix += token;
        } else {
            int p = priority[token]; // Get priority of the new operator
            while ( !s.empty() && priority[s.top()] >= p ) { // lazy evaluation; left first
                postfix += s.top(); // pop if top operator is not less important than new op
                s.pop();
            }
            s.push(token);
        }
    }
    while (!s.empty()) {postfix += s.top(); s.pop();}
    return postfix;
}

```

Lazy Evaluation

Setting Priority of Operators

- $\wedge$ represents power 7
- $\wedge$ before $+$ $-$ $*$ $/$
- $*$ $/$ before $+$ $-$
- $*$ same as $/$
- $+$ same as $-$

```
map<char,int> priority =
    {{'+',3},{'-',3},{'*',5},{ '/',5},{'^',7}};
```

Complications with Parentheses

- In parentheses is like sub-expression
- Open parentheses always push (very high priority)
- Close parentheses is very low priority
- If found, pop until sees open parentheses

input 2 * (3 + 4)

read	output	stack
2		
*	2	
(	2	*
3	2	(*
+	2 3	(*
4	2 3	+ (*
)	2 3 4	+ (* (pop all)
	2 3 4 + *	

```
int p = priority[token];
while ( !s.empty() && priority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
s.push(token);

int p = outpriority[token];
while ( !s.empty() && inpriority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
if (token == ')') s.pop(); else s.push(token);
```

Priority of Operators with Parentheses

	+	-	*	/	$\wedge$	(	)
In expression	3	3	5	5	7	9	1
In stack	3	3	5	5	7	0	

Right to Left Operator

	+	-	*	/	$\wedge$	(	)
In expression	2	2	4	4	8	9	1
In stack	3	3	5	5	7	0	

Note Power in expression is more important than in stack

input 2 + 3 - 4 ^ 5 ^ 6

read	output	stack
2		
+	2	
3	2	+
-	2 3	+
4	2 3 +	-
^	2 3 + 4	^ -
5	2 3 + 4	^ -
^	2 3 + 4 5	^ -
6	2 3 + 4 5	^ ^ -
	2 3 + 4 5 6	^ ^ -
	2 3 + 4 5 6 ^ ^ -	

Conclusion

- Implementations of stack to solve problems
- Main operations: push / pop / top
- Make stack with array
- If reserve enough space, all runs is $\Theta(1)$; constant time

Lecture 6

Queue

- Just like a normal queue in real life.
 - First-in-First-Out data structure
 - One way in (back of queue), one way out (front of queue)
 - push = add data to the back of the queue
 - pop = remove data from the head of the queue

Basic

```
#include <iostream>
#include <queue>
#include <vector>

using namespace std;

int main() {
    queue<int> q;
    q.push(1);
    q.push(2);
    q.push(3);
    while (q.empty() == false) {
        cout << q.front() << endl;
        q.pop();
    }
    cout << "-- example 2 --" << endl;
    queue<vector<int>> q2;
    vector<int> v1 = {1,2,3};
    vector<int> v2 = {99,88,-1};
    q2.push( v1 );
```

```

q2.push( v2 );
cout << q2.back()[1] << endl;
cout << q2.front().size() << endl;
auto x = q2.front();
q2.pop();
cout << x[0] << endl;
}

size_t    q.size()
bool      q.empty()
void      q.push(T data)
void      q.pop()
T          q.front()
T          q.back()

```

Limitation

- Same limitation as stack
 - No iterator
 - No `begin()`, `end()`
 - Can only access front and back of the queue
 - If we wish to access all members, we have to pop it all
 - Do not call `front()`, `back()`, `pop()` when the queue is empty

Radix Sort

Queue Application: Fast sorting with no comparison

Overview

- Put all data in an array
- For each digit X , from LSD to MSD
 - PUT TO QUEUE step: Sort by digit X by putting all of data from the array into B queues
 - B is the base of the number
 - For example, for a base 10 number, we will have Queue[0] to Queue[9]
 - Put into the queue labelled with that digit
 - GET FROM QUEUE step: Start from queue 0 to queue $B - 1$, remove data from the queue and put back to the array

Example

Input 115 15 42 305 21 8 463

Round 1 (digit 0), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
					305				
					15				
	21	42	463		115			8	

⇒ 21 42 463 115 15 305 8

Data is now sorted by the last digits, because we pop from queue 0 to 9

Round 2 (digit 1), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
8	15								
305	115	21		42		463			

⇒ 305 8 115 15 21 42 463

Data is now sorted by last two digits, because we goes from queue 0 to 9, which is grouped by digit 1 and the data in each queue is sorted by the last digit.

Round 3 (digit 2), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
42									
21									
15									
8	115		305	463					

⇒ 8 15 21 42 115 305 463

Code

```
#define base 10

int getDigit(int v, int k) {
    // return the kth digit of v (MSD is digit 0)
    int i;
    for (int i=0; i<k; i++) v /= base;
    return v % base;
}

// d = number of digits
void radixSort(vector<int> &data, int d) {
    queue<int> q[base];
    for (int k=0; k<d; k++) {
        for (auto &x : data) {
            q[getDigit(x,k)].push(x);
            for (int i=0, j=0; i<base; i++) {
                while(!q[i].empty()) {
                    data[j++] = q[i].front(); q[i].pop();
                }
            }
        }
    }
}
```

Breadth First Search

Queue Application: Gotta generate 'em all

The Problem

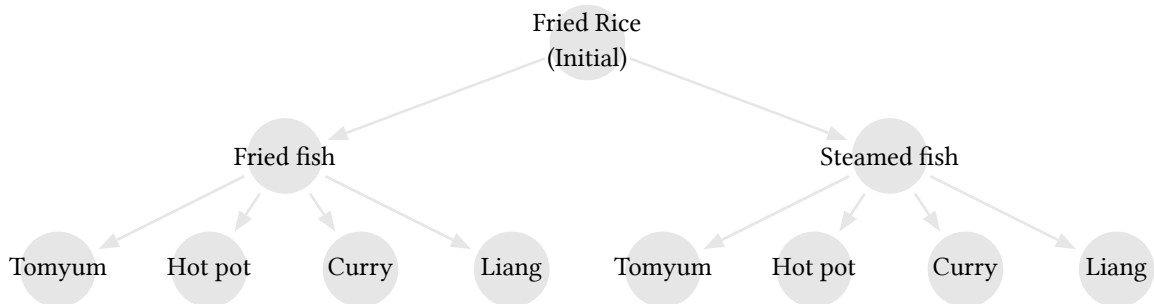
- Given a positive integer X
- Start with a number 1, find a sequence of arithmetics operations, either “ $\times$ 3”, or “ $/$ 2” that makes 1 into X
 - the $/$ 2 is integer division, e.g., $5 / 3 = 1$ (not 1.6666)
 - The sequence must be as short as possible
- Example
 - $10 = 1 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2$
 - $31 = 1 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2 / 2 / 2 \times 3 \times 3 / 2$

The Idea

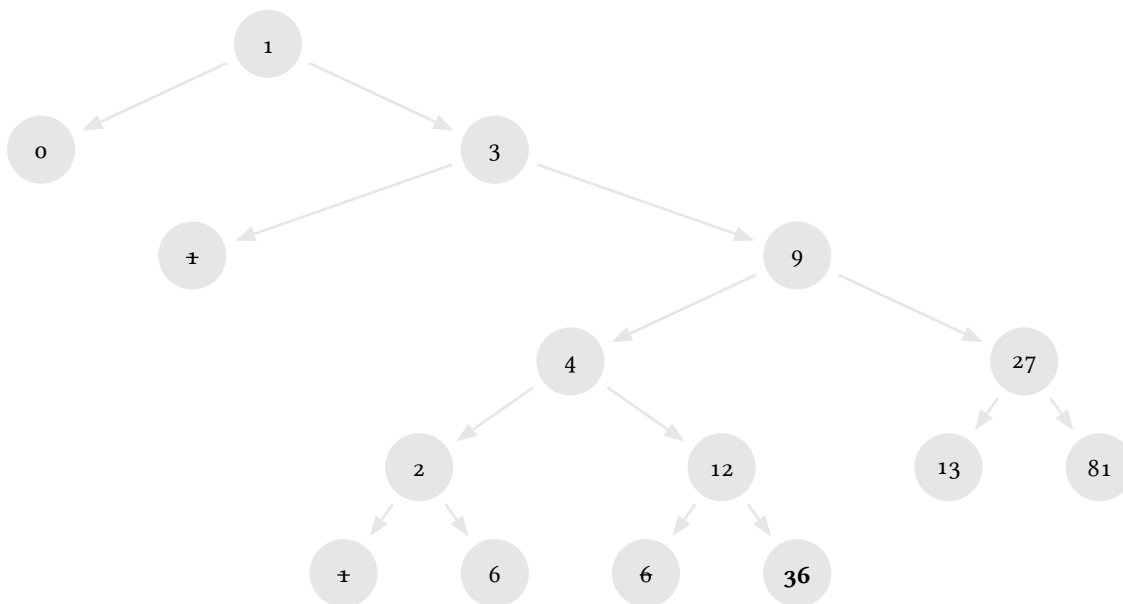
- Generate all possible sequences
 - Start with length 1, 2, 3, ... until we find one that gives X
- This is called an *exhaustive search* algorithm
 - Systematically enumerate all possible somethings

Tree Structure (Search Tree)

- A structure to illustrate *search* algorithm
- Divide into steps
 - Start with *initial solution*
 - For each possible outcome (called a *state*) of each step, *generate all* proper possible *next step*
 - Also, check if the current step is what we needed
- Written as a diagram of *node* and *edge*

Enumerate**M3D2 Problem & Solution****Back to Our Problem**

- Start with 1
- Each step is either * 3 or / 2
- Issue: might get repeated number
 - Solution: if we have found it, do not generate new step

**M3D2 Code****Code**

- `Queue` makes ordering of how we pick a state to enumerate
- From top to bottom and left to right

```

void m3d2(int target) {
    map<int, int> prev;
    queue<int> q;
  
```

```

int v = 1;
q.push(1); prev[1] = -1;
while (!q.empty()) {
    v = q.front(); q.pop();
    if (v == target) break;
    int v2 = v/2;
    int v3 = v*3;
    if (prev[v2] == 0) {q.push(v2); prev[v2] = v;}
    if (prev[v3] == 0) {q.push(v3); prev[v3] = v;}
}
if (v == target) showSolution(v, prev);
}

```

Display Solution

x	0	1	2	3	4	6	9	12	13	27	36	39	40
prev[x]	1	-1	4	1	9	2	3	4	27	9	12	13	81

showSolution

```

void showSolution(int v, map<int,int>& prev) {
    string out = "";
    while (prev[v] != -1) {
        if (prev[v] * 3 == v) {
            out = "x3" + out;
        } else {
            out = "/2" + out;
        }
        v = prev[v];
    }
    out = "1" + out;
    cout << out << endl;
}

```

⇒ 36 = 1 * 3 * 3 / 2 * 3 * 3

Lecture 7

Priority Queue

Queue with privilege

Intro

- Priority Queue is ...
 - A queue with priority
 - Item with high priority is promoted to the front of the queue
 - There is *no back of the queue*
 - Priority is defined by having more value
 - Comparison, by default, is to use operator <, i.e., if item A < B is true, then B *has higher priority*
 - We can have custom comparator
- Has the *same interface* as stack

Example

For intuitive purpose only! This is not really how priority_queue work internally

```

push(10)
< 10      >

```

```
push(30)
< 30 10    >
```

```
push(20)
< 30 20 10 >
```

```
pop()
< 20 10    >
```

```
push(15)
< 20 15 10 >
|
top
```

Basic

```
size_t      q.size()
bool        q.empty()
void        q.push()
void        q.pop()
T           q.top()
* there is no q.back()
```

Limitation

- Same limitation as stack, queue
 - *no iterator*
 - No `begin()`, `end()`
 - Use `#include <queue>`
 - Can only access `top()` of the queue
 - If we wish to access all members, we have to pop it all
 - Do not call `top()`, `pop()` when the queue is empty
- The data type must be *comparable* (similar to set and map)

Class

Quick Summary

- Syntax
 - class declaration must end with ;
 - Function definition can be outside of the class
 - Access modifier is `public:`, `private:`, `protected:`
 - Constructor is a function with the same name of the class with no return type
- Object is a variable (instantiation) of a class
 - When declared, a constructor is called

Example 1

```
#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {
```

```

        return "[" + name + " " + surname + "];";
    }
private:
    string name, surname;
};

int main() {
    Student a;
    Student b;
    a.setFullName("nattee", "niparnan");
    cout << a.getFullName() << endl;
    cout << b.getFullName() << endl;
}

```

Declaring class before writing functions

```

class Student {
public:
    void setFullName(string name, string surname);
    string getFullName();
private:
    string name,surname;
};

void Student::setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
}

string Student::getFullName() {
    return "[" + name + " " + surname + "];";
}

```

Output

```

[nattee niparnan]
[ ]

```

Example 2: Constructor

```

#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    Student(float score) {gpax = score;}
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {return "[" + name + " " + surname + "];"}
    bool is1stHonor() {return gpax >= 3.6;}
private:
    string name, surname;
    float gpax;
};

int main() {

```

```

Student a(2.95);
a.setFullName("nattee", "niparnan");

cout << a.getFullName() << endl;
if (a.is1stHonor()) {cout << "YES" endl;} else {cout << "NO" << endl;}
// Student b; // <-- cannot compile because there is no default constructor
}

```

Operator Overloading

How C++ has a function for each operator

Overview

- Let say we write `a + b` when `a` and `b` is an object of some classes
 - This can be considered the same as calling a function `plus(a,b)`
 - C++ allows us to write a function for many operator and use it as an operator
 - For example we can write a function `times(a,b)` and let it be used as `a * b`
- This is what we called operator overloading

Instead of writing

```

Matrix multiply(Matrix a1, Matrix a2) {
    multiply(a, multiply(b,c))
}

```

We want to write

```

Matrix a, b, c
a * (b * c)

```

Example

```

#include <queue>
#include <iostream>
#include <string>

using namespace std;

// use operator + symbol that will be overloaded
string operator*(string & lhs, const int & rhs) {
    string result = "";
    for (int i=0; i<rhs; i++) {
        result = result + lhs;
    }
    return result;
}

int main() {
    string a = "abc ";
    cout << (a * 3) << endl; // this gives "abc abc abc "
}

```

- Function must be named operator followed by the operator that we will overload
- Some operator takes two parameters (such as `+`, `-`, `*`, `/`, `%`)
- Some takes one (such as `++`, `--`, `!`, `*`, `&`)

Overloading less-than operator

Using with data structure that require sorting

- We have seen several data structure that requires comparability of the data, such as set, map and priority queue
- If we want to use our class with these data structure, we need to tell them how can we use them
- There are multiple ways to achieve this
 - Let us consider operator overloading

Overloading <

- As stated earlier, set, map, and priority_queue use *operator <* to compare two elements
- It does not work if we overload *operator >*

```
class Student {
public:
    Student(float score, string a, string b) {
        name = a;
        surname = b;
        gpax = score;
    }
    bool is1stHonor() {return gpax >= 3.6;}
    // not good, now our data is public
    string name, surname;
    float gpax;
    // overloading <
    bool operator<(const Student & other) const {
        return gpax < other.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    cout << (a < b) << endl; // 1
    priority_queue<Student> pq;
    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // attawith
}
```

Custom Comparator

Why custom?

- By overloading *operator <*, we have defined default ordering of that class
- What if we need another ordering, just for this *priority_queue* only
 - For example, *Student* is ordered by *gpax* by default
 - What if we want our *priority_queue* to order by name instead, while keeping the *Student* default ordering elsewhere
 - Better, can we have multiple *priority_queue* with different ordering?
- Can be done via comparator class

Example

```
#include <iostream>
#include <string>
#include <queue>
```

```

using namespace std;

class Student {...};

// Comparator class use () overloading
class StudentByNameComparator {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        return lhs.name < rhs.name;
    }
};

class GpaxThenName {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        if (lhs.gpax == rhs.gpax)
            return lhs.name < rhs.name;
        return lhs.gpax < rhs.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    Student c(4.00, "vishnu", "kotrajaras");
    cout << (a < b) << endl; // 1
    StudentByNameComparator comp1;
    GpaxThenName comp2;

    cout << comp1(a,b) << endl; // 0; can use like function

    // Use 3 template parameters
    priority_queue<Student, // class to compare
                  vector<Student>, // vector of that class
                  StudentByNameComparator> pq(comp1); // comparator class

    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // nattee

    priority_queue<Student,
                  vector<Student>,
                  GpaxThenName> pq2(comp2);

    pq2.push(a);
    pq2.push(b);
    pq2.push(c);
    cout << pq2.top().name << endl; // vishnu
}

```

Another Method, lambda-function

```

#include <iostream>
#include <string>
#include <queue>
using namespace std;

```

```
int main() {
    auto compare = [](const string& lhs, const string& rhs) {
        return lhs.size() < rhs.size();
    };

    cout << "Result of compare function = " << compare("xxx","z") << endl;

    priority_queue<string,vector<string>,decltype(compare)> pq(compare);
    pq.push("somchai");
    pq.push("z");
    pq.push("abc");
    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

Output

```
Result of compare function = 0
somchai
abc
z
```

Priority Queue Template

Templating of priority_queue

- priority_queue requires 3 template parameters
- priority_queue<T, Container = vector<T>, Compare = less<T>>
- The first one is required (which is the type of the data)
- The *second* and the *third* is optional (it has default type)
 - *Second* is the container (for now, just don't think about it)
 - *Third* is the class for comparator (the class that we use to compare)
 - This one is default to less<T>

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
    less<int> x;    // comparator that checks a < b
    greater<int> y; // checks a > b

    int a = 10;
    int b = 3;
    cout << x(a,b) << endl; // 0
    cout << y(a,b) << endl; // 1
}
```

Using Comparator for set and map

- To use custom class with set and map, we need to do the same thing, let set and map know how to sort the data
 - Either make default ordering (overload <) in the custom class
 - Or use custom comparator when declare
- For set, the declaration is set<T, Compare = less<T>>
- For map, the declaration is map<Key, T, Compare = less<Key>>

Assignment

- Is any of `vector<int>`, `set<int>`, `map<int, string>`, `queue<bool>`, `stack<vector<int>>` comparable?
 - For any class that is “YES”, how it is ordered?
 - For example, if `vector<int>` is comparable, how `{1, 2, 3}` is compared to `{1, 2, 3, 4}` or `{2, 3, 4}`

Answer

- `vector<int>` compare each element from the left; `{2, 3, 4} > {1, 2, 3, 4} > {1, 2, 3}`
- `set<int>` compare the maximum element () of each set, if equal then look the next max
- `map<int, string>` compare by Key set, then compare max of value set
- `queue<bool>` is not comparable
- `stack<vector<int>>` is not comparable

Summary**Data Structure Summary**

Data Structure	Pro	Cons	Remark
<code>pair<T1, T2></code>	Nothings... It just a pair of two data type		
<code>vector<T></code>	• Fast access [] • Fast append $\mathcal{O}(1)$	• Slow find $\mathcal{O}(N)$ • Slow insert, Slow Erase $\mathcal{O}(N)$	
<code>set<T></code>	• Fast find $\mathcal{O}(\log(N))$ • Item is sorted	• Slower to just append data than vector, stack, queue $\mathcal{O}(\log(N))$ • Iterate is also slow • Takes lots of memory $\mathcal{O}(\log(N))$	Require comparator
<code>map<Key, T></code>			• Associative data type • Also require comparator of Key_Type
<code>stack<T></code>	• Very fast push, pop $\mathcal{O}(1)$	• Very limited functionality but has special uses	• No iterator • Order of data coming out depends on something (stack, queue depends on WHEN it is pushed, pq depends on value)
<code>queue<T></code>			
<code>priority_queue<T></code>	• Fast get max • Fast delete max • Data is sorted • Memory efficient	• Slower to just append data than vector, stack, queue • Very limited functionality	• pq requires comparator

More Data Structure

- C++ has more data structure not really covered right now
 - `list` is a vector with faster input / erase but does not have fast access
 - `unordered_set`, `unordered_map` are set and map that the data is not sorted but is much faster
 - `deque` is a queue that can push, pop at both ends
 - `multiset`, `multimap` are set and map that allows duplicated entries

Lecture 8**Create Your Own CP::pair****Intro**

- We start writing our own data structure
 - In our own *namespace* (CP)
- We start with *pair*
 - Good introduction on writing a class

- Some C++ feature we need to use
 - const
 - pass-by-value, pass-by-ref
 - Header file

What should we write in a class

- Class name and namespace
- Member *variables* which define what data we want to store in the class
- Member *functions* which define behavior of the class and how member variable is manipulated
 - Include lots of special function such as *constructor*, *destructor*, *overloading*

CP:pair version 0.1 (minimal version)

- Templating
 - Parameter of a code
- No member functions
 - When we declare a pair, T1 and T2 must be declared as a type

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int,string> p1; // In p1, T1 is int, T2 is string
    CP::pair<bool,int> p2;  // In p2, T1 is bool, T2 is int
}
```

Capabilities

- Can hold 2 pieces of data
- Can use template
- Works with operator =
- Has copy constructor
- Cannot check equal
- Cannot check less than

```
#include <iostream>
#include <string>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
```

```
    cout << p2.first << "," << p2.second << endl;
}
```

```
/*
if (p1 == p2) { // won't compile
    cout << "yes" << endl;
}
```

```
if (p1 < a) { // won't compile
    cout << "yes" << endl;
}
*/
```

CP::pair version 0.2 (comparator overload)

```
#include <iostream>
#include <string>
#include <set>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // operator
        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        bool operator<(const pair<T1,T2>& other) const { // operator< must be const
            return ((first < other.first) ||
                    (first == other.first && second < other.second));
        }
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
    cout << p2.first << "," << p2.second << endl; // 20, somchai

    if (p1 == p2) {cout << "yes" << endl;} // yes
    if (p1 < a) {cout << "yes" << endl;}
    set<CP::pair<int,int>> s;
    s.insert({1,2});
    // Now we can compare and use less than (and also set, map, priority queue)
    cout << s.begin()->second << endl; // 2
}
```

Const Member

Const Parameter in Function

```
void test(int &x, const int& y) {
    x++;           // Okay; doable
    cout << y << endl; // Okay, we only read y
    for (int i=0; i<y; i++) { // Okay, too
        cout << i << endl;
    }
    y--; // ERROR: we promised NOT to change y
}
```

- Declared by putting a keyword const as a prefix
- Const parameter must not be modified inside the function
- Why? So that we know that the function does not modify the data
 - Especially when we pass-by-reference

Difference of Pass-by-reference && Pass-by-value

Pass-by-value

- The *arguments* can be either constants or variables
- Modifying *parameters* (variable inside the function) does not change the *argument's* value
- The *argument's* value is copied to the parameter
 - **SLOWER!!!** Because we have to copy

Pass-by-reference

- The *arguments* must be variables
- The *parameters* are the argument's variables
 - Modify the *parameter* also modify the *argument's* variable
- **Faster**

Const Member Function

```
class ccc {
public:
    int a,b;

    void inspect() const { // This function promises NOT to change anything
        if (a < b) cout << "yes" << endl; // Okay

        // b += 20; // <-- NOT OKAY
    }

    void mutate() { // This function might change something
        if (a < b) a += 10; // Okay
    }
};

void test2(ccc& changeable, const ccc& unchangeable) {
    changeable.inspect();
    changeable.mutate();
    unchangeable.inspect();
    unchangeable.mutate(); // ERROR: attempt to change unchangeable object
}
```

- Declared by putting a keyword const after the function declaration
- Const member function cannot modify any member data
 - Also cannot call any other function that is not const

- Why? So that we know that the function does not modify its data

Constructor & List Initialization

Custom Constructor

- In STL spec of `std::pair`, it has custom constructor called *initialization constructor*

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // Custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) {
            first = a;
            second = b;
        }
        // When we have a constructor, a default constructor is not auto-generated

        // Operator
        bool operator==(const pair<T1,T2> &other) {...}
        bool operator<(const pair<T1,T2>& other) {...}
    };
}

int main() {
    CP::pair<int, bool> p(10,false);
    CP::pair<string, int> q("abc",42), r("",0);

    cout << (q < r) << endl; // 0
    priority_queue<CP::pair<string,int>> pq;
    pq.push(r);
    pq.push(q);
    cout << pq.top().first << endl; // abc

    CP::pair<string, int> x(q);
    CP::pair<string, int> y = x;

    // All below cannot be compiled
    // CP::pair<string, int> w;
    // vector<CP::pair<int, int>> v(10);
}
```

Initialization List

- Instead of writing a code to assign a value to each member, we can use *initialization list*
 - A little bit shorter code
 - Also little bit faster
 - Only way to init *const* member

```
namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;
```



```

        // custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

Default Constructor

- A constructor used when we simply declare an object
- Auto-generated by init all members with its default constructor
- Won't be auto-gen if we have any other constructor

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
        T1 first;
        T2 second;

        // Constructor
        pair() : first(), second() { }
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

Include & Header file

Include File

- Usually, our data structure (which is written as a class) will be used by several programs in several files.
 - It is better NOT TO copy our data structure code into each file
- Rather, put our code in a file and include it where it is needed
 - Better if we want to change it
 - Better compilation
- Introducing ".h" files

C++ header file (.h) and #include

- To put a content of one file into another file, we use #include "filename" keyword
- C++ will simply put the content of *filename* into where we #include it
- #include has more benefit
 - Separation of *declaration* (what it is) and *definition* (how it works)
 - Not really explored in this class
- We usually use .h for a file that we will include but that is not a rule, we can use other extension

Problem with include

a.h

```

class a {
    int m1, m2;
};

```

b.h

```

#include "a.h"

```

```

class b {
    a x;
};

```

c.h

main.cpp (expanded)

```

class a {
    int m1, m2;
};

```

```

class b {
    a x;
};

```

```

// There are multiple copies of class a
class a {
    int m1, m2;
};

```

```
#include "a.h"
```

```
class c {
    a y;
};
```

main.cpp

```
#include "b.h"
#include "c.h"
```

```
int main() {
    b b1;
    c c1;
}
```

Fix problem

a.h

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif
```

b.h

```
#include "a.h"
```

```
class b {
    a x;
};
```

c.h

```
#include "a.h"
```

```
class c {
    a y;
};
```

```
class c {
    a y;
};
```

```
int main() {
    b b1;
    c c1;
}
```

main.cpp (expanded)

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif

class b {
    a x;
};

// class a here is taken out of
// compilation because A_H is defined
#ifndef A_H
// #define A_H
// class a {
//     int m1, m2;
// };
#endif

class c {
    a y;
};

int main() {
    b b1;
    c c1;
}
```

Summary

Major

- **Templating**: allow user to write a code that works with different data type
- **Constructor**: a function called when we create a variable
 - Default constructor
- **Operator overloading**: for less-than and equality
- pass-by-ref vs pass-by-value

Minor

- Header file
- const
- List initialization

Exercise (no grader)

1. Modify operator< so that it compare *second* before *first*
2. Modify operator< so that when we call sort(v.begin(), v.end()) where w is a vector of our pair, it is sorted from *Max* to *Min*
3. Write operator!= and operator>=

Final version CP.h

```
#ifndef _CP_PAIR_INCLUDED_
#define _CP_PAIR_INCLUDED_

namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;

        pair() : first(), second() {}

        pair(const T1 &a, const T2 &b) : first(a), second(b) { }

        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        // 1.
        bool operator<(const pair<T1,T2>& other) const {
            return ((second < other.second) ||
                    (second == other.second && first < other.first));
        }

        // 2.
        bool operator<(const pair<T1,T2>& other) const {
            return ((first > other.first) ||
                    (first == other.first && second > other.second));
        }

        // 3.
        bool operator!=(const pair<T1,T2>& other) const {
            return !(*this == other);
        }

        bool operator>=(const pair<T1,T2>& other) const {
            return !(*this < other);
        }

    };
}
#endif
```

Lecture 9

CP::vector

Our first “real” data structure

Intro

- Now we will create more complex data structure `CP::vector`
- It can store variable length array
 - Implemented as a dynamic array
- Can be accessed by operator `[]`
 - Additional operator to be overloaded
- We also have to create our own iterator
 - Implemented as a pointer

Key Idea

- Vector stored 3 things (3 member data)
 - `mData`: A dynamic array, large enough space to store current data and might have reserved space
 - `mSize`: Number of data stored
 - `mCap`: Size of the dynamic array (maybe large than `mSize`)
- If the dynamic array is full and more data is being added, we create a new dynamic array and relocate data to the new array
 - This is called `expand`
 - Each `expansion` takes very long time
- Dilemma
 - Large reserve = less often relocation but use more memory
 - Small reserve = more frequent relocation but less memory

Example

```
mSize  mCap      mData
  4      5    [10, 3, 1, 5, ]
```

```
push_back(9)
```

```
  5      5    [10, 3, 1, 5, 9]
```

```
push_back(4)
```

```
  6     10    [10, 3, 1, 5, 9, 4, , , , ]
```

How much reserve should we have?

- Whenever we need to relocate, we `double` the capacity that we currently have
- If we start with a vector with zero size and continuously add data one by one, for example by `push_back`
 - Each expansion will take time `equal` to `the number of data` when we relocate (this is very slow)
- By doubling the size every time we expand, we can show that on average, `each addition of a data` (such as `push_back`, `insert`) `takes constant time!!!`

Pointer

How to implement a dynamic array (and also iterator)

Pointer & Memory

- Each variable is some block in computer memory
 - Programming language just map our `variable name` to that `block of memory`
 - Programming language works with the address of that block
- Pointer variable is a variable that stor `address of memory`

- ▶ Pointer needs type i.e., address of int is not the same as address of bool
- ▶ We can use operator & to ask for the **address** of a **variable**
- ▶ We can use operator * to ask for the **data** of an **address**

Example

```
int x, y;    // this is normal variable x and y
int *a;     // this is int pointer variable a
int *b;     // another int pointer

x = 10;     // say x is at address 3267
a = &x;     // a points to x
b = a;      // b also points to x
*b = 30;    // change the address value to 30
y = *b;     // set value of y to 30 (hard copy)

cout << &x << endl;    // 0x7ffee22885ac
cout << &y << endl;    // 0x7ffee22885a8 +4
cout << &a << endl;    // 0x7ffee22885a0 +8
cout << &b << endl;    // 0x7ffee2288598 +8
cout << sizeof(int) << endl;    // 4 bytes
cout << sizeof(*int) << endl;    // 8 bytes
```

Pointer Arithmetic

- Pointer can be added, subtracted by integer
 - ▶ It **moves the address by the size of the type** of the pointer
 - ▶ For example, when X is an int (which is 4 bytes) X + 10 result in an address 40 bytes away from X
- Two pointers of the same type can be subtracted
 - ▶ The result is the address **difference divided by size of the type** of the pointer

```
int main() {
    bool x, y, z;
    x = 1; y = 2; z = 3;
    bool *a, *b;

    a = &x;
    b = a+2;

    cout << "&x = " << &x << endl;    // &x = 0x6afed4
    cout << "&y = " << &y << endl;    // &y = 0x6afed8
    cout << "&z = " << &z << endl;    // &z = 0x6afedc
    cout << " a = " << a << endl;    // a = 0x6afed4
    cout << " b = " << b << endl;    // b = 0x6afedc
    cout << b-a << endl;            // 2
}
```

Dynamic Array

- **Dynamic Array** variable of type T is a **pointer** to the **starting address** of consecutive block of type T
- Let A be a dynamic array of int
 - ▶ A[x] refer to the x^{th} block starting from A
- Static array works in the same way, that's why accessing A[x] is very fast for an array
 - ▶ It just refers to the address A[x] is $*(A + x * \text{sizeof}(T))$

new and delete operator

- For a datatype T, new T *allocates* a block with a size of T and then call a *constructor* of T, return the address of that memory

- For a datatype T, `new T[n]` allocates n blocks of T and then call a *constructor* of T in each block, return the address of the first block
 - For example, we use `new int [10]` to create a dynamic int array of 10 elements
- For a pointer X, `delete X` calls the *destructor* and then *de-allocates* memory pointed by X
- For a dynamic array X, `delete []` calls the *destructor* of all blocks in the dynamic array and *de-allocates* the memory allocated by X

Example

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    test *a, *b;    // pointers of test

    a = new test;    // created new test pointed by a
    a->data = 10;    // (*a).data = 10
    cout << a->data << endl;    // 10
    delete a;        // destroyed

    b = new test[4]; // created 4x
    b[0].data = 10;
    b[1].data = 20;
    b[2].data = 30;
    b[3].data = 40;
    delete [] b;    // deleted 4x
}
```

Memory Leak

- For everything, this is created by new, we must call delete on it
- If you do not, that memory is not deleted until all memory is used up

```
#include <iostream>
using namespace std;

void leaked() {
    int *a;
    a = new int[2000];
}

int main() {
    for (int i=0; i<1000000; i++) {
        int *a;
        cout << i << endl;
        leaked();
    }
}

// terminate called after throwing an instance of 'std::bad_alloc'
```

Smart Pointer (NOT A SUBJECT OF THIS COURSE)

- A better way is to use C++ smart pointer `shared_ptr(T)`
- Smart pointer is a pointer that can delete itself when it go out of scope

- Similar concept to [Java Garbage Collection](#)
- Still possible to have memory leak

vector.h

Version 0.1

- Start with vector that can do push_back, pop_back, and []
- Also with custom constructor

```
namespace CP {
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {...}
    void expand(size_t capacity) {...}
    void ensureCapacity(size_t capacity) {...}
public:
    vector() {...}
    vector(size_t capacity) {...}
    ~vector() {...}
    // access
    T& at(int index) {...}
    T& operator[](int index) {...}
    // modifier
    void push_back(const T& element) {...}
    void pop_back() {...}
};
}
```

Basic Constructor

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

public:
    vector() {
        int cap = 1;
        mData = new T[cap]();
        mCap = cap;
        mSize = 0;
    }

    vector(size_t cap) {
        mData = new T[cap]();
        mCap = cap;
        mSize = cap;
    }
}
```

```
vector<int> v;
```

```
vector<int> w(5);
```

mSize	mCap	mData		mSize	mCap	mData
0	1	[]		5	5	[0,0,0,0,0]

Destructor

- Since we new mData, we have to delete it
 - Or face a memory leak problem

```
~vector() {
    delete [] mData;
}
};
```

Object Life Cycle

- Normal object
 - Object is created (constructor called) when declared
 - Object is destroyed (destructor called) when go out of scope
- Object created by new (both new T or new T[]) Object is created when new Object is destroyed when delete

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    cout << "-- Life cycle --" << endl;
    cout << "- normal cycle -" << endl;
    test u;
    u.data = 99;
    for (int i=0; i<5; i++) {
        test t;
        t.data = i*10;
    }
}
```

Output

```
-- Life cycle --
- normal cycle -
created      <-- u
created
0 destroyed
created
10 destroyed
created
20 destroyed
created
30 destroyed
created
40 destroyed
99 destroyed <-- u
```


Accessing Data

- The return type is T& which is a reference
- Same deal as pass-by-reference, this is called **return-by-reference**
 - So we can do `v[i] = 30` or `v[i]++`
- What is returned is actually that variable
- Also notice the difference between `at()` and `operator[]`

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {
        if (n < 0 || (size_t)n >= mSize) {
            throw std::out_of_range("index out of range");
        }
    }

public:
    T& at(int index) {
        rangeCheck(index);
        return mData[index];
    }

    T& operator[](int index) {
        return mData[index]; // return the variable
    }
};
```

Add, remove data

- `push_back` first check if we have reserved space
 - If not, we expand
- Then, the data is put to `mData[mSize]`
- Removing Data is done by just reduce the size

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    void expand(size_t capacity) {
        T *arr = new T[capacity](); // create a new dynamic array
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[i]; // move all data
        }
        delete [] mData;
        mData = arr; // delete old data and point to new one
        mCap = capacity;
    }

    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap; // double the size
            expand(s);
        }
    }
};
```

```

    }
}
public:
    void push_back(const T& element) {
        ensureCapacity(mSize+1);
        mData[mSize++] = element; // add size and put element
    }
    void pop_back() {
        mSize--;
    }
};

```

Vector Constructors

Problem of vo.1

- Copy constructor and assignment operator is incorrect
 - It is auto generate to copy all variables (but not the data it points to)
- Rule of three in c++
 - Consider destructor, copy constructor, assignment operator
 - If any of them is written in the code, we mostly need all of them
- Since c++11, it's rule of four an a half

```

int main() {
    CP::vector<int> w(5);

    for (int i=0; i<5; i++) w[i] = i*10;
    CP::vector<int> x(w); // this creates shallow copy of w
    CP::vector<int> y = w;
    x[3] = -1; // changes both w and y
    cout << y[3] << endl; // -1
    cout << w[3] << endl; // -1
}

```

vo.2, add small access functions

- empty and size also exists in other data structure
- size_t is non-negative integer type

```

bool empty() const {
    return mSize == 0;
}

size_t size() const {
    return mSize;
}

size_t capacity() const {
    return mCap;
}

```

vo.2 adding copy constructor & assignment operator

```

// copy constructor
vector(const vector<T>& a) {
    mData = new T[a.capacity()]();
    mCap = a.capacity(); // copies only value
    mSize = a.size();
    for (size_t i=0; i<a.size(); i++) {
        mData[i] = a[i];
    }
}

```

```

    }
}

// copy assignment operator
vector<T>& operator=(vector<T> &other) {
    // protect against self-destruct
    if (mData != other.mData) {
        // delete current data
        delete [] mData;
        // copy the new data
        mData = new T[other.capacity()]();
        mCap = other.capacity();
        mSize = other.size();
        for (size_t i=0; i<a.size(); i++) {
            mData[i] = a[i];
        }
    }
}

```

Copy-and-swap

- Utilize written copy-constructor and destructor
- Shorter code

```

// copy assignment operator using copy-and-swap idiom
vector<T>& operator=(vector<T> other) { // notice the pass-by-value!!!
    // other is copy-constructed which will be destruct at the end of this scope
    // we swap the content of this class to the other class and let it be destructed
    using std::swap;
    swap(this->mSize, other.mSize);
    swap(this->mCap, other.mCap);
    swap(this->mData, other.mData);
    return *this;
}

```

Iterator, Insert, Erase, & Analysis

vo.3 Iterator and typedef keyword

```

template <typename T>
class vector {

protected:
    T *mData;
    size_t mCap;
    size_t mSize;
protected:
    typedef T* iterator; // let iterator be type name of T pointer

    // iterator
    iterator begin() {
        return &mData[0];
    }

    iterator end() {
        return begin()+mSize;
    }
}

```

- See that pointer works just like how `std::vector::iterator` works
- In fact, iterator is actually a pointer
- typedef keyword allows us to map a type name
 - `CP::vector<int>::iterator` is `int*`
 - `CP::vector<bool>::iterator` is `bool*`

insert

- `push_back` actually call `insert(end(), element)`
- Question: why we need `pos`?

```
iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize+1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1];
    }
    mData[pos] = element;
    mSize++;
    return begin()+pos;
}

void push_back(const T& element) {
    insert(end(), element);
}
```

erase

- See that both `insert` and `erase` also change `mSize`

```
void erase(iterator it) {
    while((it+1)!=end()) {
        *it = *(it+1);
        it++;
    }
    mSize--;
}
```

Exercise

- Read the following function and see how it works in `vector.h`
 - `resize`, `clear`
 - non-stl function
 - `insert_by_pos`
 - `erase_by_pos`
 - `erase_by_value`
 - `contains`
 - `index_of`
- Read in [here](#)

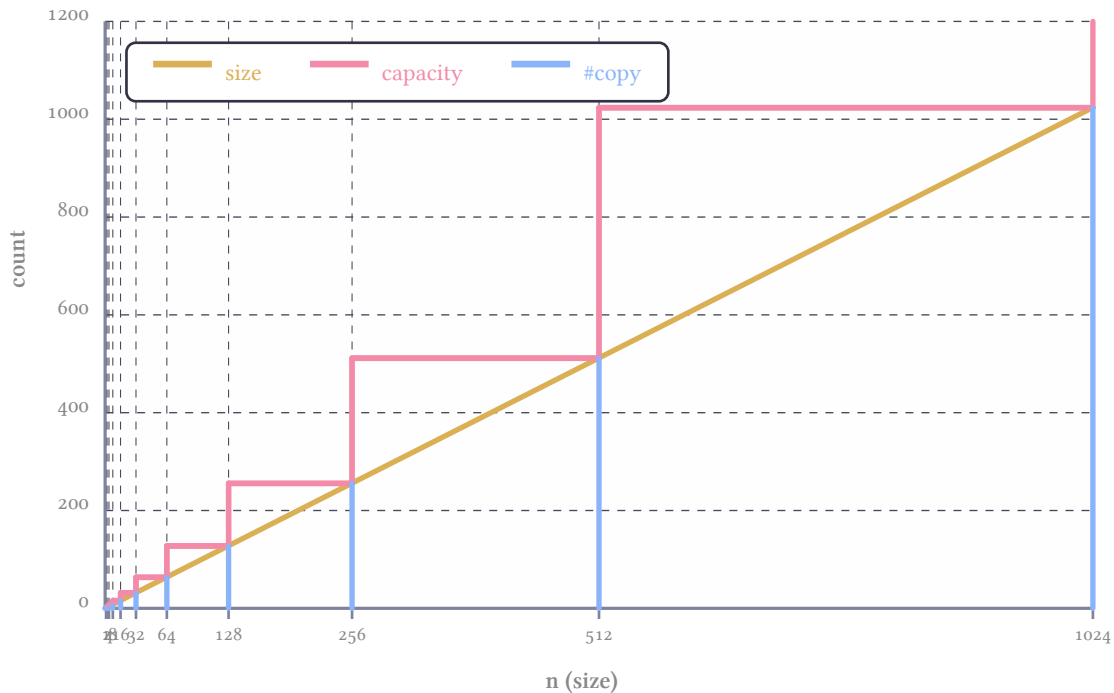
Will do this part later

Analysis of how many data is copied by `push_back`

- When full, `push_back` have to move all data to a new dynamic array
- `ensureCapacity` double the size

Size and Capa & Copy Count

- How much copy we need?



Lecture 10

Stack

Intro

- Now we will create less complex data structure `CP::stack`
- Just like a vector without iterator, insert, erase, resize, at end operator[]
 - Add `top()` which is just a shorthand of looking at the last element
- That's it, really

Key Idea

- The data is stored in the same way as a vector
 - The first element of `mData` is the bottom of stack while the last element is the top of the stack
- We just take `vector.h` and remove unnecessary function

```

| | stack
| |
| A | mSize mCap mData
| X | 4 5 [A, B, X, A]
| B |
| A |
|__|

```

stack.h

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        void expand(size_t capacity) {...}
    };
}

```

```

    void ensureCapacity(size_t capacity) {...}
public:
    // constructor
    stack(const stack<T>& a) {...}
    stack() {...}
    stack<T>& operator= {...}
    ~stack() {...}
    // capacity operation
    bool empty() const {...}
    size_t size() const {...}
    // access
    const T& top() const {
        return mData[mSize-1];
    }
    // modifier
    void push(const T& operator) {...} // This is push_back
    void pop() {...} // This is pop_back
};
}

```

Speed of each operation

- All read operation always take constant time
 - `size()`, `top()` simply return something that is directly accessible
- All modify operation also take constant time
 - `push()` is constant on average (same as `push_back` of vector)
 - `pop()` is always constant

Stack by Vector

- Instead of writing our own function, there is another way to write a stack
- We simply use `vector` as our sole data member
- Benefit: `code reuse`
- Drawback: `almost none` except that we need one more layer of function call

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        vector<T> v;
    public:
        // default constructor
        stack() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& top() const { return v.back(); }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.pop_back(); }
    };
}

```

Lecture 11

CP::queue

Will the circle be unbroken?

Intro

- Queue, unlike stack, require more sophisticated technique to achieve fast performance
- We start by writing a simple class that just work (slowly)
- Then we try to improve it

Key Idea

- Just like stack, we will use the same format as vector, using dynamic array to store data
- However we have to somehow manage how we works with `front()` and `back()` of the queue

vo.1 simple implementation of the queue

- To illustrate this idea, we will use a **vector** as our data member
- `push(e)` is simply `v.push_back(e)`, this is fast
- `front()` is `v[0]`, `back()` is `v[v.size()-1]`, this is also fast
- `pop()` is `v.erase(v.begin())`, this is slow (always proportional to `v.size()`)
 - Unlike `std::queue` which has very fast `pop()`

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
    public:
        // default constructor
        queue() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& front() const { return v[0]; }
        const T& back() const { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.erase(v.begin()); }
    };
}
```

vo.1 example

```
CP::queue<int> q;      q  v
                        mSize mCap mData
                        0      1  ----> [ ]

q.push(10);            q  v
                        mSize mCap mData
                        1      1  ----> [10]

q.push(20);            q  v
                        mSize mCap mData
                        2      2  ----> [10, 20]

q.push(30);            q  v
                        mSize mCap mData
                        3      4  ----> [10, 20, 30, ]

q.pop();               q  v
                        /      /
```

```

mSize  mCap  mData    /    /
  2      4      ----> [20, 30, , ]

```

vo.2 faster queue

- Add more data member mFront, initialized as 0
- push(e) is simply v.push_back(e), this is fast
- front() is v[mFront], back() is v[v.size()-1], this is also fast
- pop() is mFront++, this is fast
 - However, we don't really remove anything when pop

```
#include <vector>
```

```

namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
        int mFront;
    public:
        // constructor
        queue() : v(), mFront() {}
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size() - mFront; }
        // access
        const T& front() const { return v[mFront]; }
        const T& back() { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { mFront++; }
    };
}

```

vo.2 example

```

CP::queue<int> q;      q      v
                        mFront mSize mCap mData
                        0      0      1      ----> [ ]

q.push(10);            q      v
                        mFront mSize mCap mData
                        0      1      1      ----> [10]

q.push(20);            q      v
                        mFront mSize mCap mData
                        0      2      2      ----> [10, 20]

q.push(30);            q      v
                        mFront mSize mCap mData
                        0      3      4      ----> [10, 20, 30, ]

q.pop();               q      v
                        mFront mSize mCap mData
                        1      3      4      ----> [10, 20, 30, ]

```

Problem with vo.2

- Fast but **use too many space**
- Queue grows according to **how many time push is called**
 - regardless of how many pop is called
- The data stored in the vector can be **much larger** than the actual data in the queue
- Does not really work in real world


```
for (int i=0; i<1000000; i++) {
    q.push(i);
    q.pop();
}
std::cout << q.size() << std::endl;
```

Circular Queue

Final Idea

mFront	mSize	mCap	mData	*
2	2	4	---->	[, , 30, 40]
q.push(50);				
mFront	mSize	mCap	mData	*
2	3	4	---->	[50, , 30, 40]
q.push(60);				
mFront	mSize	mCap	mData	*
2	4	4	---->	[50, 60, 30, 40]
q.pop();				
mFront	mSize	mCap	mData	*
3	3	4	---->	[50, 60, , 40]
q.push(99);				
mFront	mSize	mCap	mData	*
3	4	4	---->	[50, 60, 99, 40]
q.push(1);				
mFront	mSize	mCap	mData	* \ \ \
0	5	4	---->	[40, 50, 60, 99, 1, , ,]

- We take vo.2 and **reuse** the area at the beginning of mData
 - Expand when necessary
 - Re-arrange when expand

Circular Queue

- We can think of mData to be circular
 - End of the last element of the mData is connected to the first element
- Consider i^{th} element
 - the next element is $(i+1) \% \text{mCap}$
 - The previous element is $(i-1+\text{mCap}) \% \text{mCap}$
 - Next k element is $(i+k) \% \text{mCap}$

Circular Queue Implementation

queue.h

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        size_t mFront;
        void expand(size_t capacity) {...}
        void ensureCapacity(size_t capacity) {...}
    public:
        // constructor - almost the same but have to take care of mFront
        queue(const queue<T>& a) {...}
        queue() {...}
    };
}
```

```

    queue<T>& operator=(queue<T> other) {...}
    ~queue() {...}
    // capacity function - same as vector
    bool empty() const {...}
    size_t size() const {...}
    // access - circular queue implementation
    const T& front() const {...}
    const T& back() const {...}
    // modifier
    void push(const T& element) {...}
    void pop() {...}
};
}

```

Ctor, Dtor, copy

```

template <typename T>
class queue {
protected:
    T *mData;   size_t mCap;   size_t mSize;   size_t mFront;
public:
    // default constructor
    queue() : mData(new T[1]()), mCap(1),
             mSize(0), mFront(0) { }
    // copy constructor
    queue(const queue<T>& a) : mData(new T[a.mCap]()), mCap(a.mCap),
                             mSize(a.mSize), mFront(a.mFront) {
        for (size_t i = 0; i < a.mCap; i++) {
            mData[i] = a.mData[i];
        }
    }
    // copy assignment operator
    queue<T>& operator=(queue<T> other) {
        using std::swap;
        swap(mSize, other.mSize);
        swap(mCap, other.mCap);
        swap(mData, other.mData);
        swap(mFront, other.mFront);
        return *this;
    }
    ~queue() {
        delete [] mData;
    }
};

```

front(), back(), pop()

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
public:
    // access
    const T& front() const {
        return mData[mFront];
    }
};

```

```

    }
    const T& back() const {
        return mData[(mFront + mSize - 1) % mCap];
    }
    // modify
    void pop() {
        mFront = (mFront + 1) % mCap;
        mSize--;
    }
};

```

- back = mFront + mSize - 1
 - Also circular (by % mCap)
- pop = move mFront by 1
 - Also circular
 - Also change size

push, expand

- push add data to (mFront + mSize) % mCap
 - The space just after back()
- Expand re-pack the mData so that mFront is 0
- ensureCapacity is the same

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
    void expand(size_t capacity) {
        T *arr = new T[capacity]();
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[(mFront + i) % mCap];
        }
        delete [] mData;
        mData = arr;
        mCap = capacity;
        mFront = 0;
    }
    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
            expand(s);
        }
    }
public:
    void push(const T& element) {
        ensureCapacity(mSize+1);
        mData[(mFront + mSize) % mCap] = element;
        mSize++;
    }
};

```

Analysis

- All access, modification is fast (constant time)

- Space is re-used
 - It is not shrunk when mSize reduce
 - Space is not more than double of maximum mSize during its lifetime

Queue Exercise

- We implement circular queue by maintain mFront and use circular logic ($\% \text{ mCap}$) to calculate the position of back of the queue
 - Can we maintain mBack instead?
 - Can we maintain both mFront and mBack but not mSize?
- How about mCap, if we know mFront, mSize, mBack, can we calculate mCap?

mFront	mSize	mBack	front()	back()	size()
YES	YES	No	$v[\text{mFront}]$	$v[(\text{mFront} + \text{mSize}) \% \text{ mCap}]$	$\text{mSize}()$
No	YES	YES	????	$v[\text{mBack}]$	mSize
YES	No	YES	$v[\text{mFront}]$	$v[\text{mBack}]$	????

Answer

- Use $\text{front}() = v[(\text{mCap} + \text{mBack} - \text{mSize}) \% \text{ mCap}]$
- Use $\text{size}() = (\text{mCap} + \text{mBack} - \text{mFront}) \% \text{ mCap}$, but this will break when $\text{mBack} == \text{mFront}$ because it can't determine whether the queue is empty or full.
- Impossible, mCap can be scaled to any size larger than the current elements.

Now, meet deque

- Can you modify queue to include
 - $\text{push_front}()$, add to the front of the queue
 - $\text{pop_back}()$, remove from back of the queue
- All operation should still be constant time

Answer

```
void push_front(const T& element) {
    ensureCapacity(mSize+1);
    mFront = (mCap + mFront - 1) % mCap;
    mData[mFront] = element;
    mSize++;
}

void pop_back() {
    mSize--;
}
```

Lecture 12

Complexity Analysis

How fast is our code?

Overview

- Introducing a **measurement** of efficiency of a program
- Why we need measurement
- How we measure
 - How to analyze our code to get a measurement
- What is measurement
 - **Asymtotic Notation**

Key Idea

- We need a useful way to describe efficiency of our code
 - Useful = able to be **easily use to predict** how much resource (time / memory) that our program will need
 - Useful = **not overly complex** in analysis
 - Need to balance between usefulness and complexity
- Ultimately, we introduce a **class of efficiency** that says how our code use resource with respect to size of data
 - Focus on **on growth of resource usage**

Preview

```
int find_max(vector<int> v) {
    int m = v[0];
    for (size_t i = 0; i < v.size(); i++)
        if (v[i] > m)
            m = v[i];
    return m;
}
```

- This code takes time **directly proportional** to **the size of the data**
 - Size N takes time T
 - Size $5N$ should take time $5T$

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

- This code takes time **directly proportional** to **the square of the size of the data**
 - Size N takes time T
 - Size $5N$ should take time $25T$

Why don't we use real world clock?

- Ultimately, we want to know **how long our program takes** to do each operation
 - Use in design, how much resource we need
 - Help use **choose appropriate data structure**
- **Real world clock measurement is possible** but has **many drawback**
 - System dependency
 - Too complex (we have to build our system)
 - Too specific

Dependency of System

- Same program in different system = different time
 - CPU
 - RAM
 - Compiler
 - Operating parameter
 - Heat? Other running program?

How to measure

- Counting instruction
 - Depend on code only
- Dependency on size of data
 - Focus on large data

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

// 1
 // $2 + n + n$ ($n = v.size()$)
 // $(2 + n + n) * n$
 // $2 * n * n$
 // $1 * ????$ ($\leq (d)$)
 // 1

```
}
```

```
// Total = 1 + 2+n+n + (2+n+n)n + 2n^2 + n^2 + 1
//          = 4 + 4n + 5n^2
```

Time per instruction

- In reality, different CPU instructions use different time
- Same instruction but different CPU also use different number of cycle
- **However, we just ignore it**
 - For now

Focusing on large data

- In many cases, the size of the data we are working with will affect the time our code use
- Large data usually mean longer time
- What matter is when the data is large

Growth rate simplifies analysis

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = i+1; j < v.size(); j++)
            if (v[i] + v[j] == k)
                count++;
    return count;
}
```

// i=0: 1 + n-0 + n-1
 // i=1: 1 + n-1 + n-2
 // <-- i=2: 1 + n-2 + n-3
 // . . .
 // i=n-1: 1 + 1 + 0
 // sum = $\Sigma(1+2n-2i-1) = n^2 + n$

```
// Total = 1 + 2+n+n + n^2+n + n^2 + n^2 + 1
//          = 4 + 3n + 3n^2
```

Small Detail

n	$5n^2 + 4n + 4$		$3n^2 + 3n + 4$		n^2	
	raw	growth	raw	growth	raw	growth
10	544		334		100	
20	2084	3.830 88	1264	3.784 43	400	4
40	8164	3.917 47	4924	3.895 57	1600	4
80	32 324	3.959 33	19 444	3.948 82	6400	4
160	128 644	3.979 83	77 284	3.9747	25 600	4
320	513 284	3.989 96	308 164	3.987 42	102 400	4
640	2 050 564	3.994 99	1 230 724	3.993 73	409 600	4
1280	8 197 124	3.9975	4 919 044	3.996 87	1 638 400	4
2560	32 778 244	3.998 75	19 668 484	3.998 44	6 553 600	4
5120	1.31×10^8	3.999 37	78 658 564	3.999 22	26 214 400	4
10 240	5.24×10^8	3.999 69	3.15×10^8	3.999 61	1.05×10^8	4
20 480	2.1×10^9	3.999 84	1.26×10^9	3.9998	4.19×10^8	4

When counting instruction, it is usually OK to focus on **most executed line**

Measurement by Growth Rate

What?

Why?

- Growth rate = how much resource usage growth with respect to change of input
 - Resource usage = number of instruction used
 - Input = size of data
- Emphasizes long term trend
- System independent
 - The result can be used to predict behavior on any system
- Focus on change of resource usage with respect to size of input
- Can regard small detail
 - Simple to calculate
 - Applicable in real world

Asymtotic Notation

Classification of growth rate

Overview

- Formally, it is a **set of function** having the **growth rate** related to something
- The definition focus **on growth of the function** while **disregard small detail**
- Also provide some workaround on dependency of value of input

What is?

- A kind notation written as $O(f(n))$
 - O can be one of $\mathcal{O}, \Theta, \Omega, \sigma, \omega$
 - $f(n)$ is some expression
- Example $\mathcal{O}(n)$ or $\Theta(n^2 + 3)$ or $\omega(n^2 \log(n))$
- Usage
 - “This code is $\mathcal{O}(n)$ ” (read as Big-Oh of n)
 - “This function takes time in $\Theta(\log(n))$ ” (read as Big-theta of log n)
 - “Time complexity of this program is $\mathcal{O}(n^2)$ ”

Meaning

- “A is $\Theta(f(x))$ ” means the **growth rate of A** is **equal** to the **growth rate of $f(x)$**
- “B is $\mathcal{O}(f(x))$ ” means the **growth rate of B** is **less than or equal** to the **growth rate of $f(x)$**
- For Ω, σ, ω , (**Big-Omega**, **little-oh**, **little-omega**), we won’t use it for now but the meaning is similar (which are **more than or equal**, **less than**, **more than**, respectively)
- Conventionally, we usually use **N for the size of the data**

Usage

- Let **$f(n)$** be the number of instruction needed by **code A** when the **size of data is n**
- We will calculate asymptotic notation that **$f(n)$** is a member of
 - Find $g(n)$ such that $f(n)$ is $\mathcal{O}(g(n))$ or $\Theta(g(n))$ (or $\Omega(g(n))$ or ...)
- Let’s say we have analyze that **$f(n)$ is $\mathcal{O}(g(n))$**
 - We now understand that the growth rate of instruction required by n grows slower or the same as how $g(n)$ grow

Comparing growth rate of $f(n)$ and $g(n)$

- The relation of growth rate of $f(n)$ and $g(n)$ depends on the value of $f(n)/g(n)$ when $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \begin{cases} 0 & \text{if } f(n) \text{ grows slower than } g(n) \\ c & \text{if } f(n) \text{ grows similarly to } g(n) \\ \infty & \text{if } f(n) \text{ grows faster than } g(n) \end{cases}$$

Example

- $f(n) = 4 + 3n + 4n^2$
- $g(n) = n^2$

$$\lim_{n \rightarrow \infty} \frac{4n^2 + 3n + 4}{n^2} = \lim_{n \rightarrow \infty} \left(4 + \frac{3}{n} + \frac{4}{n^2} \right) = 4$$

- Hence $f(n)$ grows similarly to $g(n)$
 - Therefore, $f(n) = \Theta(n^2)$

Another Example

- $f(n) = 0.00005n^2$
- $g(n) = 100000n$

$$\lim_{n \rightarrow \infty} \frac{0.00005n^2}{100000n} = \lim_{n \rightarrow \infty} 10^{-10}n = \infty$$

- Hence $f(n)$ grows faster than $g(n)$ (also means $g(n)$ grows slower than $f(n)$)
 - Therefore $g(n) = \mathcal{O}(n^2)$
 - and $f(n) = \Omega(100000n)$

Dependency on the value of input

- Consider `vector::insert(iterator it, T value)`
- The time it takes depends on both the size of the vector and the value of it
 - Larger size $\rightarrow$ more time
 - Closer to `end()` $\rightarrow$ less time

Big-O describes upper bound

- `vector::insert` is $\mathcal{O}(n)$
 - Its growth rate does not exceed n
 - There are cases that it may grow less than n (insert at `end()`)
- This is very useful in real world
 - Knowing maximum load
 - Not overly complex in analysis

Big-O Example

- Find is $\mathcal{O}(n)$
 - At worst, it can't find value and a & b points to `begin()` & `end()`
 - This case, find growth rate is n
 - At best, it always find value at the first position (a)
 - This case, find growth as 1

```
bool find(iterator a, iterator b, T value) {
    while (a < b) {
        if (*a == value)
            return true;
        a++;
    }
    return false;
}
```

L'Hôpital's Rule

$$\lim_{n \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{n \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

If $f(x), g(x)$ must be diffable, $g(x)$ is non-zero, $\lim_{n \rightarrow \infty} f'/g'$ exists

- Can help
- $F(n) = \log n$
- $g(n) = n^{0.5}$

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{\log(n)}{\sqrt{n}} &= \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(10)\sqrt{n}} \\
&= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\ln(n)}{\sqrt{n}} \\
&= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} \\
&= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n}} \\
&= 0
\end{aligned}$$

Exercise

- $f(n) = \lg^c n$
- $g(n) = n^k$
- We know that $c > 0, k > 0$
- Does $f(n)$ grow slower than $g(n)$?

Will do this part later

Big Theta**Big-Theta is tight bound**

- `std::count` always go through entire array
- Regardless of the value in the array, it always perform `if (*first == value)`

```

size_t count(iterator first, iterator last, const T& value) {
    size_t ret = 0;           // 1
    for (; first != last; ++first) { // n + n
        if (*first == value)    // 1 * n
            ret++;              // 1 * ??? ( <= (d) )
    }
    return ret;               // 1
}

```

```

// Total = 1 + n+n + 1*n + 1*n + 1
//         = 2 + 4n

```

More Example

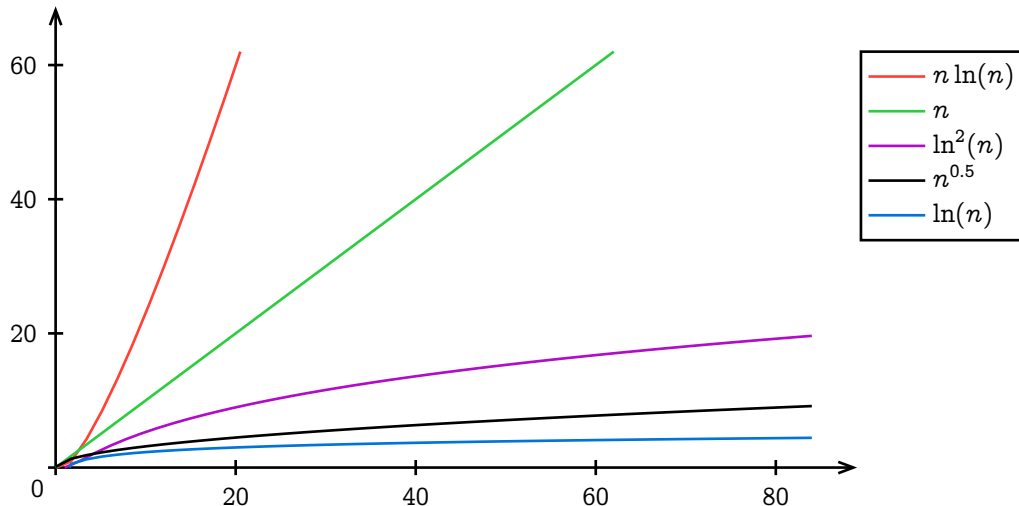
$\Theta(1)$	$\Theta(n)$	$\Theta(n^2)$
$f(n) = 5$	$f(n) = n$	$f(n) = n^2$
$f(n) = 0$	$f(n) = n + 3$	$f(n) = \binom{n}{2} = \frac{n(n-1)}{2}$
$f(n) = c$	$f(n) = \frac{n}{1200} + 86$	$f(n) = 400n^2 + an + b$
	$f(n) = 40000000n$	

Observation: multiplicative and addition constants in $f(n)$ can usually be ignored, since it will be disregarded by $\lim$, Degree cannot.

Well known growth rate class

$\Theta(1)$	Constant
$\Theta(\log(n))$	Logarithm
$\Theta(\log^c(n)), c \geq 1$	Polylogarithm
$\Theta(n^a), 0 < a < 1$	Sublinear

$\Theta(n)$	Linear
$\Theta(n \log(n))$	Linearithmic
$\Theta(n^2)$	Quadratic
$\Theta(n^c), c \geq 1$	Polynomial
$\Theta(c^n), c > 1$	Exponential
$\Theta(n!)$	Factorial



$\ln^2(n)$ grows slower than $n^{0.5}$. At around $n = 56000$, $\ln^2(n)$ is less than $n^{0.5}$.
 x^4 grows slower than 2^n . At around $n = 16$, x^4 is less than 2^n .

Beware

- It is **wrong** to say that `vector::insert` is $\Theta(n)$
 - Because there is a case that it grows slower than n
- It is **wrong** to say that `vector::push_back` is $\Theta(n)$
 - Because there is a case that it grows slower than n
- It is **ok** to say that `std::count` is $\mathcal{O}(n)$
 - Because while it always grows as n , it **does not grow faster than n**
 - $\mathcal{O}(n)$ is upper bound
 - But it is better to say that `std::count` is $\Theta(n)$

How to analyze using asymptotic notation

- Write a code
- Calculate the function $f(n)$ that counts the number of instruction of the code when the data is size n
 - Usually, just focus on **most executed line**
- Find $g(n)$ and a notation X such that $f(n)$ is $X(g(n))$
 - It's either **Big-O** or **Big-Theta**. If there is a case that it can grow slower than $G(n)$, use Big-O

Another Example

- Let's analyze `vector::push_back`

```

iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize + 1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1]; // <----- Most executed line: 1 (b)
    }
    mData[pos] = element;
    mSize++;
}

```

```

    return begin()+pos;
}
void push_back(const T& element) {
    insert(end(), element);
}

void expand(size_t capacity) {
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i]; // <----- Most executed line: 0, 0, 0, 0, 2, 4, 8, 16 (a)
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}

void ensureCapacity(size_t capacity) {
    if (capacity > mCap) {
        size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
        expand(s);
    }
}

// (a) can be 0 to n
// (b) can also be 0 to n
// Best case 0 + 0
// Worst case n + n
// vector::push_back is O(n)

```

Another Definition for $\mathcal{O}$ and Θ

- Using set builder notation
- $\mathcal{O}(g(n)) = \{f(n) \mid \text{there exists } c > 0 \text{ and } n_0 \geq 0 \text{ such that } f(n) \leq cg(n) \text{ for } n \geq n_0\}$
- $\Theta(g(n)) = \{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- The result is the same as definition using lim

Summary

- We use Asymtotic Notation to describe efficiency of a program
 - Measure instruction count instead of time
 - Focus on growth rate of instruction count
- Find most frequently executed line in the code and count it
- Maps to Big-Theta if we have tight bound
- Use Big-O if we have upper bound

Example

Showing $\log_2(n!)$ is $\Theta(n \log_2(n))$

- Will use set definition of Big-Theta
 - $\{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- Need to find c_1, c_2 and n_0

Finding c_1 and c_2

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1$$

$$n! \leq n \times n \times n \times n \times \dots \times n$$

$$\log n! \leq \log n^n \quad \boxed{\log n^n = n \log n \text{ when } n \geq 1}$$

Found $c_2 = 1$ for upper bound $\log n! \leq n \log n$ when $n \geq 1$

$$n! = n \times (n-1) \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times \left(\left\lfloor \frac{n}{2} \right\rfloor - 1 \right) \times \dots \times 1$$

$$n! \geq \left\lfloor \frac{n}{2} \right\rfloor \times \left\lfloor \frac{n}{2} \right\rfloor \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times 1 \times \dots \times 1$$

$$n! \geq \left(\frac{n}{2} \right)^{n/2} \quad \log \left(\frac{n}{2} \right)^{n/2} = \frac{n}{2} \log n - \frac{n}{2}$$

$$\log n! \geq \log \left(\frac{n}{2} \right)^{n/2} \quad 0.5n \log n - 0.5n \geq 0.4n \log n \text{ when } n \geq 32$$

Found $c_1 = 0.4$ for lower bound $\log n! \geq 0.4n \log n$ when $n \geq 32$

$$0.4n \log n \leq \log n! \leq n \log n \text{ when } n \geq 32, c_1 = 0.4, c_2 = 1, n_0 = 32$$

$$\Rightarrow \log n! \text{ is } \Theta(n \log n) \text{ and } n \log n \text{ is } \Theta(\log n!)$$

Lecture 13

Priority Queue Simple Implementation

Featuring Binary Heap

Overview

- Simple implementation of `priority_queue`
- Quick intro to **Graph** and **Tree**
- **Binary Heap**
- `priority_queue` with **Binary Heap**

`priority_queue`

- Queue by value
- Max-in-First-Out

```
int main() {
    priority_queue<int> pq;
    pq.push(4);
    pq.push(20);
    pq.push(3);

    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

Vo.1, `priority_queue` by vector

- Use **vector** to store data
- push = simply `push_back`
- top, pop = find the max value and return / erase
- `max_element` returns iterator to max element

```
namespace CP {
    template <typename T>
    class priority_queue {
    protected:
        std::vector<T> v;
    public:
        bool empty(); { return v.empty(); }
        bool size(); { return v.size(); }
```

```

    void push(const T &e) { v.push_back(e); }

    T top() { return *std::max_element(v.begin(), v.end()); }

    void pop() { v.erase(std::max_element(v.begin(), end())); }
};
}

```

max_element

```

iterator max_element(iterator first, iterator last) {
    if (first == last) return last;
    iterator largest = first;
    ++first;
    for (; first != last; ++first)
        if (*largest < *first)
            largest = first;
    return largest;
}

```

Vo.1 complexities

push	$\mathcal{O}(1)$ *amortized
top()	$\Theta(n)$
pop()	$\Theta(n)$

Vo.2 faster pop, top (and push??)

- vo.1 has many drawbacks
 - Consecutive call of top is slow (it shouldn't)
 - Both pop and top works almost the same
- vo.2 focus on **slower push** while keeps **pop, top fast**
- Make the array **sorted**
 - Max will be at the back
 - Fast pop, top

```

namespace CP {
    template <typename T>
    class priority_queue {
    protected:
        std::vector<T> v;
    public:
        bool empty(); { return v.empty(); }
        bool size(); { return v.size(); }

        T& top() { return v[v.size()-1]; }

        void pop() { v.erase(v.end()-1); }

        void push(const T &e) {
            // do something
        }
    };
}

```

vo.2 push

- Maintain that the vector is sorted at every push

```

void push(const T& e) {
    v.push_back(e); // <--  $\mathcal{O}(1)$ 
    sort(v.begin(), v.end()); // <--  $\mathcal{O}(n \log(n))$ 
}

void push(const T& e) {
    auto it = v.begin();
    while (it < v.end() && *it <= e) // <--  $\mathcal{O}(n)$ 
        it++;
    v.insert(it, e); // <--  $\mathcal{O}(n)$ 
} // Total of this program is  $\mathcal{O}(n)$ 

void push(const T& e) { // upper_bound is  $\mathcal{O}(\log(n))$ 
    v.insert(std::upper_bound(v.begin(), v.end(), e), e); // <--  $\mathcal{O}(n)$ 
}

```

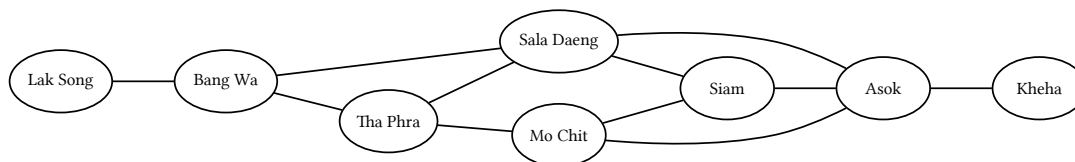
Which one is better?

- vo.1 fast push
- vo.2 fast pop, top
- Depends on which operation we use most often
- The real version works like vo.2, we maintain some **rules** of the data that is stored in the vector such that
 - We know where max is (for fast pop)
 - Much faster push, a little bit slower pop
 - Use structure called **Binary Heap**

Graph & Tree

Graph

- Discrete Math Graph
- A math model that describe entities and connectivity between them



Graph Model

- Graph consists of two things
 - **Nodes (vertex, vertices)** are **things** we want to connect
 - **Edges** are pairs, each pair is a **connectivity** between two nodes
- Graph $G = (V, E)$ where V is a set of nodes and E is a set of edges

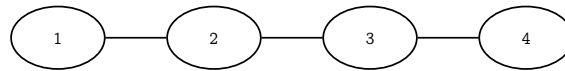
$V = \{\text{"Mo Chit", "Siam", "Asok", "Sala Daeng", "Tha Phra", "Lak Song", "Bang Wa", "Kheha"}\}$

$E =$

$\{(\text{"Mo Chit", "Asok"}), (\text{"Mo Chit", "Siam"}), (\text{"Tha Phra", "Mo Chit"}), (\text{"Sala Daeng", "Tha Phra"}), \dots\}$

Tree

- A special kind of graph
 - Has N nodes and $N - 1$ edges
 - Every nodes must be connected (we can start from any node and can walk through edge to reach any node)

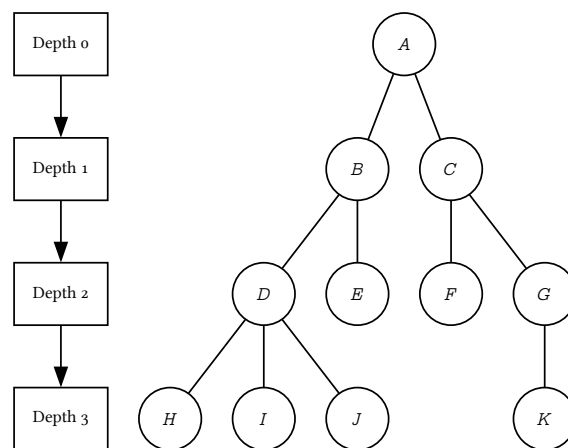


$$V = \{1, 2, 3, 4\}$$

$$E = \{(1, 2), (2, 3), (3, 4)\}$$

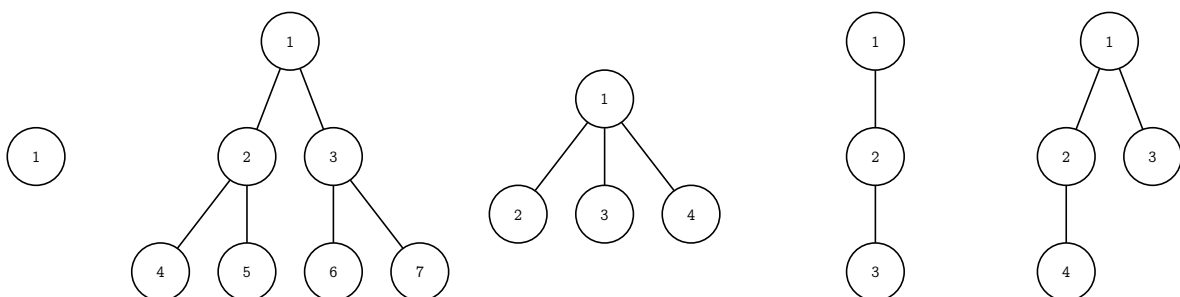
Rooted Tree

- Tree where one node is defined as **root**
- For an edge in a rooted tree, a node that is closer to the root is called **parent** while the other node is called **child**
- **Ancestor** of node A = parent of parent ... of A
- **Descendant** of node A = child of child ... of A
- Root is usually drawn at the top and is considered as the **starting point**
 - Root is at level 0 (depth 0)
 - Children of root are drawn at the same level at **level 1** (depth 1)
 - Children of children of root are at **level 2** (depth 2)



Complete Binary Tree

- Binary Tree = a tree that every node has at most 2 children
- Complete tree = the tree must be filled with every possible node at every level (except the deepest level which must be filled as far to the left as possible)
 - Blank tree is considered a complete binary tree



OK

OK

Not binary

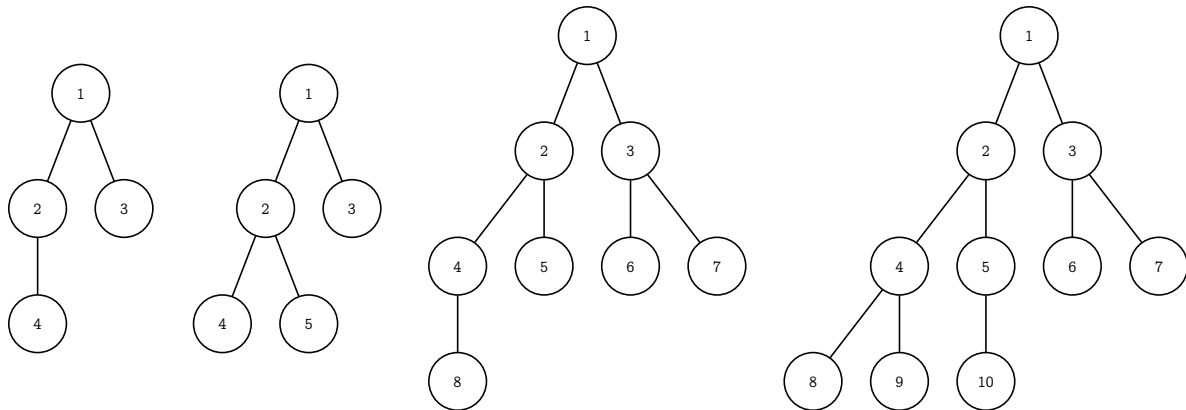
Not complete (at depth 1)

OK

Exercise

- Draw a Complete Binary Tree that has 4, 5, 8, 10 nodes

Hint: The answer is unique (There are exactly 1 way to draw a complete binary tree of size k)

**Special Property of a Complete Binary Tree**

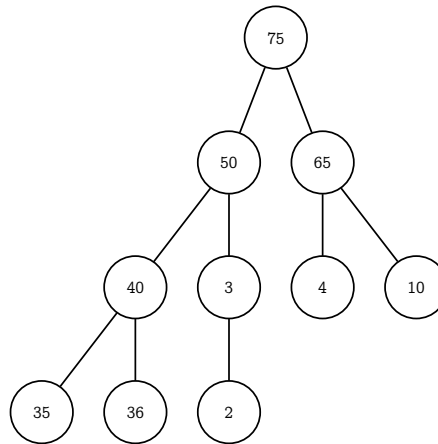
- There is exactly one way to go from any node to any node
- Maximum depth is $\log_2 n$ where n is the number of nodes
 - Because we require completeness, and we have 2 possible children

Binary Heap

Use Complete Binary Tree to make `priority_queue`

**Binary Heap**

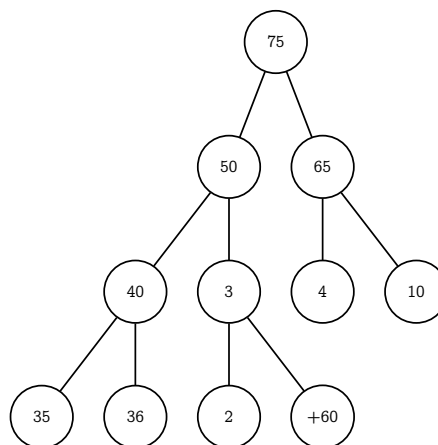
- We use **Complete Binary Tree** to store data
 - A **value** is stored at the **node**
- When data is modified (via push or pop), we must maintain these rules
 1. Tree must always be **Complete Binary Tree**
 2. For any node, its **value must be greater or equal that of its children**



Adding data to Binary Heap

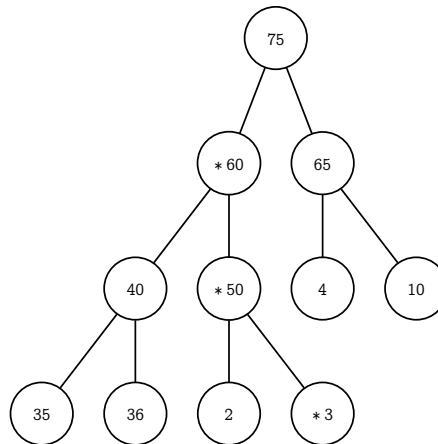
- Maintain Binary Heapness
 - structure of data
 - value of data
- Structure rules says where the new nodes should be
 - Next to **right-most child** of **deepest level**
 - But if we put new data there, the **value rules might be broken**
 - Fix it

push(60)



Fix from adding a new node

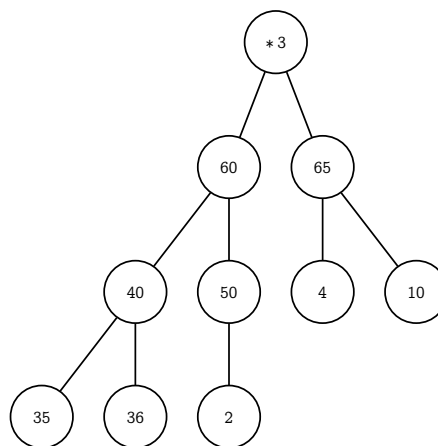
- Fix:
 - Check where we just add a node, if value rules is broken, swap with parent
 - After swap, re-check with new parent
 - Kepp doing until correct or at root



See that: after each swap, the swapped nodes does not violate with **its new children**

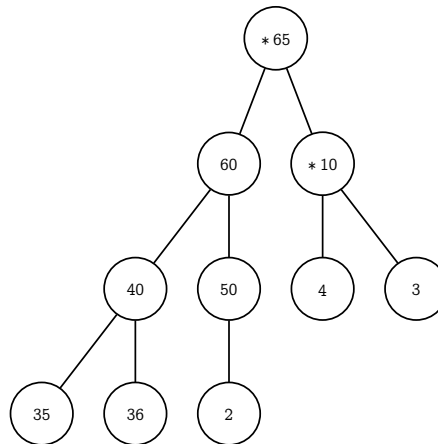
Delete maximum data

- Similar to push, we will try to maintain structure first
- delete will remove root, find something to replace
 - Use the lowest, right-most node
- Value rules might be broken
 - Fix it



Fix from deleting root node

- Fix:
 - Start at replaced root, if value rules is broken, swap with maximum child
 - After swap, re-check with new children
 - Beware! There is a case where we might have only one child
 - Keep doing until correct or has no children



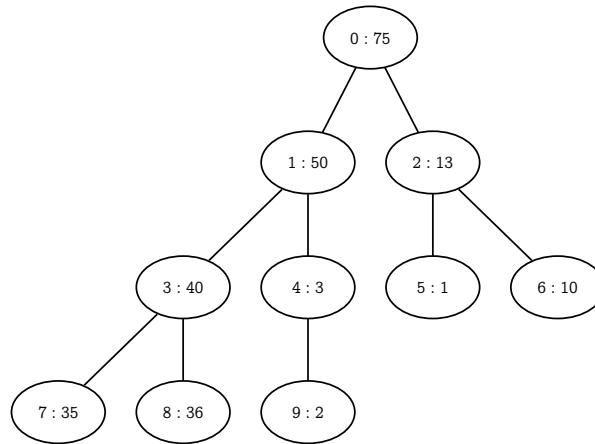
Analysis

- How fast is push, pop
- push
 - Add to vector is $\mathcal{O}(1)$ amortized
 - Fixing value rules is $\mathcal{O}(h)$ where h is the maximum number of depth of the tree (we call this value tree height)
 - Notice that tree height is $\mathcal{O}(\log(n))$
- pop
 - Fixing value rules is $\mathcal{O}(h)$ where h is the maximum number of tree height

Push + Fix Up

How to store a tree?

- Use **dynamic array**
 - Each node can be labelled from 0 to $n-1$
- **Root** is at 0
- **Left child** of node i is at $(i*2)+1$
- **Right child** of node i is at $(i*2)+2$
- **Parent** of node i is at $(i-1)/2$



Dynamic array: 0 1 2 3 4 5 6 7 8 9
 [75, 50, 13, 40, 3, 1, 10, 35, 36, 2]

Layout

```

template <typename T, typename Comp = std::less<T>>
class priority_queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    Comp mLess;
    void expand(size_t capacity) {} // Fix value rules
    void fixUp(size_t idx) {}
    void fixDown(size_t idx) {}
public:
    // constructor
    priority_queue(priority_queue<T,Comp>& a);
    priority_queue(const Comp& c = Comp());
    priority_queue<T, Comp>& operator=(priority_queue<T,Comp> other);
    ~priority_queue();
    // capacity function
    bool empty() const;
    size_t size() const;
    // access
    const T& top();
    // modifier
    void push(const T& element);
    void pop();
};
  
```

Constructor

- Using `list initialize`
- See that `mData` is dynamic array in the same way as vector
- `mLess` is something that is just either copied or default initialize
 - Will talk about it later

```

priority_queue(const Comp& c = Comp()):
    mData( new T[1]() ),
    mCap( 1 ),
    mSize( 0 ),
    mLess( c )
{ }

priority_queue(priority_queue<T, Comp>& a) :
    mData(new T[a.mCap]()),
    mCap(a.mCap),
    mSize(a.mSize),
    mLess(a.mLess)
{
    for (size_t i = 0; i < a.mCap; i++) {
        mData[i] = a.mData[i];
    }
}

```

Destructor and Copy Assignment Operator

- Using standard copy-and-swap idiom

```

~priority_queue() {
    delete [] mData;
}

priority_queue<T,Comp>& operator=(priority_queue<T,Comp> other) {
    using std::swap;
    swap(mSize,other.mSize);
    swap(mCap,other.mCap);
    swap(mData,other.mData);
    swap(mLess,other.mLess);
    return *this;
}

```

Push

- See that the **right-most child of the deepest level** is at `mData[mSize-1]` and the new node should be at `mData[mSize]`
- We do the same thing as vector's `push_back`
- Then fix the value rule

```

void expand(size_t capacity) { // Same as CP::vector
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i];
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}

void push(const T& element) {
    if (mSize + 1 > mCap)
        expand(mCap * 2);
    mData[mSize] = element;
    mSize++;
    fixUp(mSize-1);
}

```

Fix Up

- Instead of actual swap, we perform insert and find appropriate position at the same time

```
void fixUp(size_t idx) {
    T tmp = mData[idx];
    while (idx > 0) {
        size_t p = (idx - 1) / 2;
        if ( tmp < mData[p] ) break;
        mData[idx] = mData[p];
        idx = p;
    }
    mData[idx] = tmp;
}

mData  p = /, idx = *, tmp = 60
  0   1   2   3   4   5   6   7   8   9  10
[ 75, 50, 13, 40, /3,  1, 10, 35, 36,  2,*60]

[ 75,/50, 13, 40, *3,  1, 10, 35, 36,  2,  3]

[/75,*50, 13, 40, 50,  1, 10, 35, 36,  2,  3]

tmp = 60 < 75 = mData[p] => break;mData[idx] = tmp;
[ 75, 60, 13, 40, 50,  1, 10, 35, 36,  2,  3]
```

mLess

- priority_queue allows a custom comparator
- Custom comparator X
 - We must be able to $X(a, b)$ where X will compare a and b and return true only when a is less than b
 - X is a **variable** that implement operator()
- Initialize at constructor as variable mLess to be of type Comp in template
- Any comparison of our data (type T) must be done by mLess

```
void fixUp(size_t idx) {
    T tmp = mData[idx];
    while (idx > 0) {
        size_t p = (idx - 1) / 2;
        if ( mLess(tmp, mData[p]) ) break;
        mData[idx] = mData[p];
        idx = p;
    }
    mData[idx] = tmp;
}
```

Pop + FixDown**Pop**

```
void pop() {
    mData[0] = mData[mSize-1];
    mSize--;
    fixDown(0);
}
```

```
void fixDown(size_t idx) {
    T tmp = mData[idx];
```

```

size_t c;
while ((c = 2 * idx + 1) < mSize) {
    if ( c + 1 < mSize && mLess(mData[c], mData[c+1]) ) c++;
    if ( mLess(mData[c], tmp) ) break;
    mData[idx] = mData[c];
    idx = c;
}
mData[idx] = tmp;
}

```

- while loop checks if we have **at least one child**
- c is the index of **highest value child**
 - Must consider the case where we have only one child
- Exercise: read the rest yourself

Fast Construction

Construct PQ from n data

```

priority_queue(std::vector<T> &v, const Comp& c = Comp()) :
    mData( new T[v.size()]() ), mCap( v.size() ), mSize( 0 ), mLess( c ) {
    for (size_t i = 0; i < mSize; i++) push(v[i]); // n times and each log(n)
}

```

$$\text{Total} \leq \lfloor \log_2 1 \rfloor + \lfloor \log_2 2 \rfloor + \dots + \lfloor \log_2 n \rfloor$$

$$\leq \lfloor \log_2 (1 \times 2 \times 3 \times \dots \times n) \rfloor = \lfloor \log_2 n! \rfloor$$

$\lfloor \log_2 n! \rfloor \text{ is } \mathcal{O}(n \log n)$

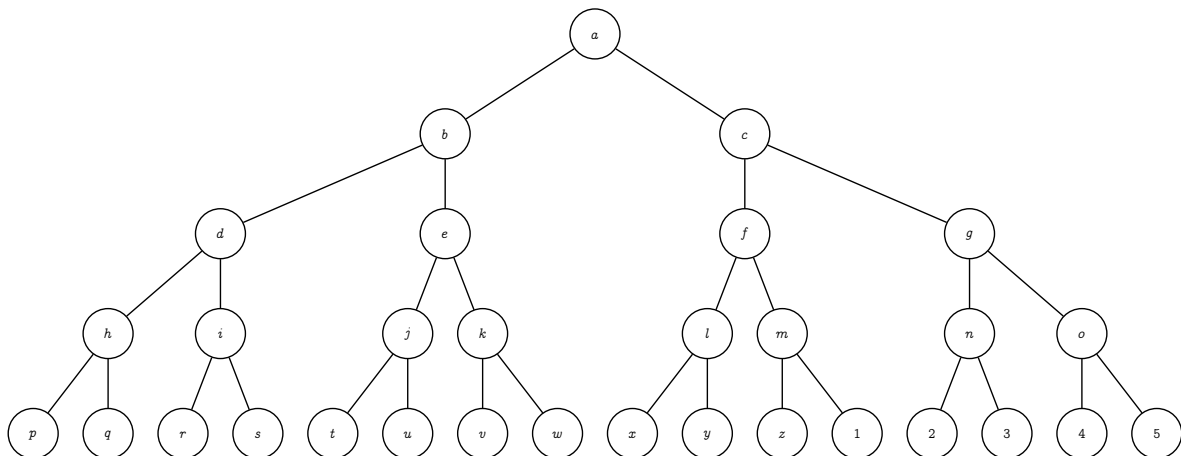
Better Method

```

priority_queue(std::vector<T> &v, const Comp& c = Comp()) :
    mData( new T[v.size()]() ), mCap( v.size() ), mSize( 0 ), mLess( c )
{
    for (size_t i = 0; i < mSize; i++) mData[i] = v[i];
    for (int i = mSize/2-1; i > 0; i++) fixDown(i);
}

```

- Consider each node to be a binary heap
- Fix down from back to front

How fast?

- Binary Tree Property:
 - There are at most 2^k nodes at depth k
 - For a tree of height h , at depth k , fix down needed at most $h - k$ iterations

Total nodes = 31

Tree height = $\log_2(31) = 4$

Depth: k , Nodes: 2^k , Max fix: $h - k$
 Total = $\sum_{k=0}^h k 2^{h-k}$

Depth	Nodes	Max fix per node
0	1	4
1	2	3
2	4	2
3	8	1
4	16	0

$$\sum_{k=0}^h k 2^{h-k} = 2^h \sum_{k=0}^h k 2^{-k} < 2^h \underbrace{\sum_{k=0}^{\infty} k 2^{-k}}_{\text{This is 2}} = 2^{h+1} = 2^{\log_2 n + 1} = \mathcal{O}(n)$$

Exercise

If we use different number of children per node, say 4-ary heap, the same logic would still holds.

1. How to keep data in dynamic array?
2. Write the new functions `fixUp` and `fixDown`
3. Is it faster?? Consider any n -ary heap

Will do this part later